Tariffs, Rules of Origin, and the Reorganization of North American Supply Chains*

Ángel García-Lascurain Fernández†

Arizona State University

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Abstract

This paper quantifies how tariffs and rules of origin drive the reorganization of North American supply chains around Mexico, and what that reorganization implies for welfare. Between 2018 and 2025 this landscape was reshaped by three shocks, which were the U.S.-China trade war, the transition from NAFTA to USMCA in 2020, and a U.S. executive order in 2025 that sharply raised tariffs on Mexican imports outside of USMCA preference. I develop a global sourcing model in which heterogeneous firms choose source countries and, product by product, whether to export under USMCA, thus having to comply with rules of origin, and calibrate it on matched census-customs microdata. Through the lens of the model, USMCA protects Mexico from U.S. tariffs, given that firms respond to a higher tariff on non-compliant goods by complying with the rules of origin rather than paying it. The value of the agreement therefore grows sharply with that tariff, meaning that eliminating it would impose a welfare loss about six times larger at the tariff level set by the 2025 order than at the tariffs in place before it. In contrast, the transition from NAFTA to USMCA was roughly welfare-neutral for Mexico, while the shift in relative sourcing efficiency across countries raised welfare.

JEL codes: F13, F14, F15, L23.

Keywords: rules of origin, USMCA, global sourcing, heterogeneous firms.

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†Department of Economics, W.P. Carey School of Business, Arizona State University, (e-mail: agarc475@asu.edu)

1 Introduction

Over the last several years North American trade policy has become increasingly complex. For nearly three decades, trade in the region operated under the North American Free Trade Agreement (NAFTA), which guaranteed frictionless commerce among the United States, Canada and Mexico as long as a sufficient share of each product was sourced within the region. Firms in turn built their continental supply chains around this certainty. This is no longer the case. Since 2018, a bilateral tariff war between the United States and China, the replacement of NAFTA by the more restrictive United States-Mexico-Canada Agreement (USMCA), and a sequence of emergency U.S. tariffs on goods that fall short of USMCA's regional content requirements have substantially altered what it costs to produce in Mexico and export to the other two nations. Preferential access can no longer be taken as given, and has instead become a changing policy margin that firms have to actively choose, comply with or forgo at every shipment.

Mexican manufacturing exports are at the center of the economy that these shocks are reorganizing. These exports amount to roughly a third of Mexican GDP, of which about 80% is destined for the United States, and are built on a long tradition of foreign investment in export-oriented plants, which today are embedded in the IMMEX program (the modern successor to the 1960s maquiladora regime). On the output side, firms face a choice on every shipment: they can either comply with USMCA's rules of origin and ship tariff-free to the U.S., or forgo preference and pay the Normal Trade Relations (NTR) tariff. Given that this choice shapes where firms source their inputs, where they sell and how much they earn, movement on the compliance margin ends up propagating through the broader economy.

The 2018 trade war raised product-specific tariffs on half of the U.S.–China trade universe. Rather than triggering reshoring, these tariffs diverted trade toward "bystander" economies, where the sectoral specialization that each country already had determined how much of this trade it was able to absorb (Fajgelbaum et al., 2024). Given its geographic proximity, deep supply-chain linkages and preferential access under NAFTA, Mexico was a natural candidate to absorb part of this trade. Between 2017 and 2025, U.S. imports from China fell by roughly half in real terms, whereas U.S. imports from Mexico grew 31%. At the same time, Mexico's own imports from China grew 32%, with intermediate and capital goods accounting for roughly 60% of that growth.[1] The scale of this restructuring can also be observed on the ground, as foreign-headquartered firms alone added roughly half a million manufacturing workers in Mexico over the same period (Utar, 2026). It is important to mention, however, that these volumes do not tell us what happened on the preferential

[1] UN Comtrade bilateral flows at HS-6, mapped to end-use categories using the UN BEC concordance.

margin. When USMCA replaced NAFTA in July 2020 with tighter rules of origin, the share of Mexico-to-U.S. trade qualifying for preference did not increase but actually fell, from 56% in 2017 to 48% in 2024. It was only after March 2025, when an executive order under the International Emergency Economic Powers Act (IEEPA) imposed an addtional 25% (additional to the existing NTR rate) tariff on Mexican imports while exempting USMCA-compliant goods, that this share jumped to about 85%. Firms had been rerouting trade through Mexico for years, but increasingly outside of the preferential regime, and it was not until the cost of staying outside of it became large enough that they formalized this trade under USMCA. While these tariffs were eventually struck down by the U.S. Supreme Court, the episode provides a glimpse at how firms react when not complying with a free trade agreement becomes prohibitively expensive. Following the ruling, the tariff on Mexican goods outside USMCA preference currently stands at 10%. It is worth pointing out that the low compliance before 2025 does not mean that the agreement mattered little to Mexico. Rather, its value remained latent while the tariff that it waives was small, and the jump in compliance after March 2025 is the firm-level counterpart of the increase in that value, which I quantify in Section 6.3.

The goal of this paper is to study the trade-off that firms face between cost-minimizing global sourcing and free-trade-agreement compliance, and to quantify its welfare implications. The relevance of this margin goes beyond what has been observed historically. USMCA is now subject to an annual joint review by the three governments, meaning that rules of origin can change from year to year. Given that continental supply chains were built under the certainty that these rules would stay in place, firms now have to revisit their compliance decision on a recurring basis instead of making it only once. The framework developed in this paper allows one to understand what each of these policy scenarios may mean for the economy.

In order to do so, I develop a global sourcing model in which heterogeneous firms combine domestic labor with domestically-sourced and imported intermediates, choose the set of countries they source from out of a global menu, and decide, product by product, whether or not to comply with NAFTA/USMCA rules of origin. Compliance grants the firm tariff-free access to the U.S. and Canadian markets, but constrains its sourcing toward North America. Opting out, on the other hand, allows the firm to source globally without any constraints at the cost of paying the NTR tariff, which is the non-discriminatory rate that a WTO member applies to imports from any other WTO member. Given that domestic value added automatically counts toward regional-content requirements, the production structure plays a direct role in the compliance margin. Firms evaluate all of their feasible sourcing strategies and only enter production if the best one yields positive profits. Aggregate income and the

domestic price level adjust in order to clear the Mexican goods market, whereas a competitive non-manufacturing sector pins down the wage and absorbs the reallocation of labor.

I calibrate the model at two different points in time using matched Mexican census-customs firm microdata: a pre-trade-war NAFTA equilibrium (2018) and a post-USMCA equilibrium (2023). The main finding of this paper is that USMCA protects Mexico from U.S. tariffs, and that it does so more the higher the tariff on non-compliant goods. With NTR tariffs in place, removing USMCA costs Mexico 1.51% of its real income, whereas at the 25-point surcharge of the 2025 order the loss amounts to 8.85%. With the preferential route available, the same surcharge costs only 0.31%. In other words, what makes the agreement valuable is the outside tariff. For the same reason, content rules cost more when non-compliance is expensive: a 20-point tightening costs about three to four times as much real income at the 2025 tariff level as at observed tariffs, which is more than the cost of the surcharge itself. The transition from NAFTA to USMCA rules, on the other hand, shows to be close to welfare-neutral, as the two-period decomposition attributes -0.03% of real income to it. When looking at the growth in real income between the two periods, the largest components that can be computed on a common panel are the growth in foreign demand ($+3.5\%$) and the shift in relative sourcing conditions ($+1.4\%$), under which imports from China rise the most. Around the 2023 baseline, however, content rules are not welfare-neutral. Tightening the regional value content (RVC) threshold by 25 points lowers compliance by 12 points and real income by 0.45%, given that most firms keep complying and buy North American content that they would not otherwise buy. Lastly, a 25% tariff on intermediates from China, a country with which Mexico holds no free trade agreement, cuts Chinese sourcing by 31% at a cost of about a tenth of a percent of real income, since the tariff rebate offsets most of the higher input costs. Other Asian sources gain the most in proportion ($+2.0\%$), while North American, European and domestic sourcing each rise by just under 1%.

This paper is related to three strands of the literature. The first is the quantitative global sourcing literature (Antràs et al., 2017, Eaton and Kortum, 2002, Head and Mayer, 2019, Head et al., 2024, Morán and Cebreros, 2025, Tintelnot, 2017), where, typically, firms differ in productivity and countries in their sourcing potential. I build on this framework in two ways. The first one is that I recover, for every firm in the matched census-customs microdata, the parameters behind its sourcing and destination decisions, which are a sourcing efficiency for each source country and a market reach for each destination. Given that these parameters are also filled in for the countries and markets that the firm does not use, both margins are able to move in the counterfactuals. The second one is that compliance is decided product by product, meaning that a multi-product firm can comply on some of its products and pay the tariff on the rest. The closest papers on this margin are Morán and Cebreros (2025)

and Head et al. (2024). The former add a NAFTA-versus-WTO choice with a fixed cost of using NAFTA to the model of Antràs et al. (2017), estimate it on Mexican customs data and target trade flows and firm profits. The latter calibrate a model of the car industry to the cost shares of North American parts and trace the Laffer curve of regional content requirements on employment. In the first paper each firm makes a single NAFTA-or-WTO choice and the sample keeps the products with a CTC rule, whereas in the second compliance is decided per car line under an RVC rule. Neither of them covers the full set of product-level rules of the treaty.

The second strand is the literature on rules of origin in preferential agreements (Conconi et al., 2018, Grossman, 1981, Ju and Krishna, 2002, Ornelas and Turner, 2024). These papers study the rules in theory or measure their effect on trade, and in each of them the rule is a single requirement. In this paper the three compliance regimes of the treaty, change in tariff classification, regional value content and the hybrid rules under which the firm can use either one, enter the same model, with their parameters attached at the HS6 level over the whole manufacturing universe of NAFTA and USMCA. This in turn allows me to take the model to firm-level data for all of manufacturing instead of a single sector.

The third strand is the literature on the reorganization of global supply chains in an environment of rising trade conflicts (Alfaro and Chor, 2025, Fajgelbaum et al., 2024, 2020, Freund et al., 2024, Gopinath et al., 2025, Grossman et al., 2023, Utar, 2026, Utar et al., 2025), which has studied how trade and production reorganize across countries, including their diversion toward Mexico. Notably, these papers do not model whether the diverted trade enters the United States under preference or not, which is the compliance friction at the center of this paper. To this literature I add a structural quantification of Mexican real income under the trade war and the NAFTA-to-USMCA transition taken together, which no structural analysis of USMCA rules of origin with an endogenous compliance margin has reported so far, to the best of my knowledge.

The remainder of this paper is organized as follows. Section 2 documents the empirical patterns in trade flows and USMCA compliance that motivate the model. Section 3 presents the structural framework. I follow by describing the calibration in Section 4 and the estimation results in Section 5. Section 6 presents the counterfactual analysis and Section 7 concludes.

2 Motivating Facts

Since 2018, North American manufacturing trade has reorganized substantially in favor of Mexico, while the share of MX→US trade claiming NAFTA/USMCA preference has moved

through three distinct regimes. There are three policy events that plausibly drive this sequence. The first took place between 2018 and 2019, when, in response to the statements of USTR (2018) on forced technology transfer and other unfair trade practices by China, the United States invoked Section 301 of the Trade Act of 1974 in order to impose tariffs on roughly half of its manufactured imports from China. The second was the replacement of NAFTA by USMCA in July 2020. USMCA raised the required North American content for about two fifths of MX→US trade, most of it in the automotive core, and left the rest unchanged or looser (Table 2). The third was the IEEPA executive order of March 2025, a 25% tariff on Mexican imports from which goods that meet USMCA rules of origin are exempt. Figure 1 shows the three trade flows and the compliance rate over this period. Section 2.1 covers the trade flows and Section 2.2 the use of USMCA preference.

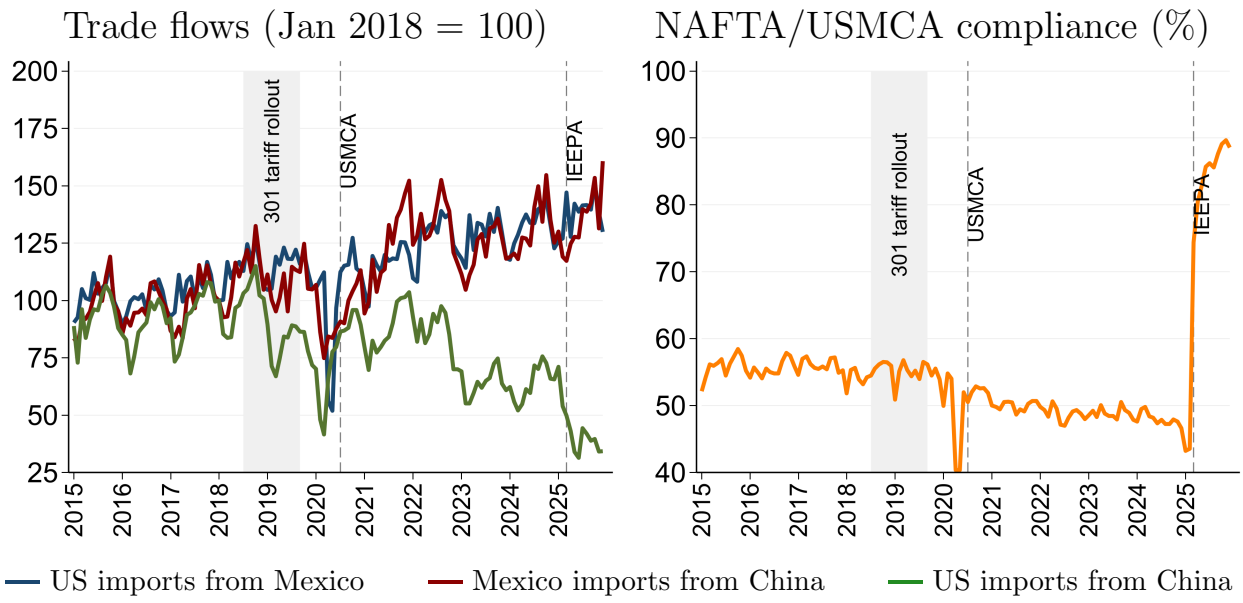


Figure 1: **Trade flows and NAFTA/USMCA compliance, 2015–2025.** Manufacturing HS6 (16–97), deflated to 2023 USD. Monthly through Dec 2025.

2.1 The Reorganization of Trade Flows

The first place where Mexico's response to the 2018 trade war can be observed is in trade volumes. U.S. imports from China fell sharply between 2017 and 2025 (−54% in real terms). Mexico took over part of that demand (MX→US: +31%), and at the same time it bought more from China itself (MX←CN: +32%). Most of these additional Chinese purchases are inputs and not consumer goods, as roughly 60% of the MX←CN growth is in intermediate and capital goods[2] and the remaining 40% in final-consumption goods.

[2]I classify goods by end use with the UN Broad Economic Categories, Rev. 5.

A natural question that arises is whether the new MX→US trade took place in the product codes that Mexico was already exporting in 2017 or in newly-emerging product lines, that is, whether the trade response occurred on the intensive or on the extensive margin. The answer shows to be overwhelmingly intensive. Table 1 splits MX→US manufacturing trade by 301 exposure and by whether Mexico already exported the product in 2017. The cell of exposed products that Mexico already exported holds 90% of 2017 trade. The two rows of new products are small. Moreover, 98% of their 2025 value falls in subheadings that only exist since the 2022 revision of the Harmonized System, which by construction have no 2017 trade and no Section 301 coding, so most of that value reflects goods that were already traded under older codes.

In the last column of Table 1 I report the trade-weighted US NTR tariff of each cell, which is the rate a Mexican exporter pays whenever it ships without claiming USMCA preference. These rates are low, at 3.3% on average for the exposed products that Mexico already exported and 3.6% across the four cells. Not claiming preference was therefore not very costly for most shipments, and as I show in Section 2.2, a large part of the eligible trade did not claim it up until the IEEPA order of March 2025, when skipping preference became much more expensive.

Table 1: MX→US manufacturing trade by 301 exposure and pre-shock exporter status, 2017–2025

		Trade share (%)		Trade value ($bn)		Avg NTR
301	Exporter status	2017	2025	2017	2025	(%)
Exposed	Existing (2017 > 0)	90.3	85.1	325.6	403.7	3.3
Exposed	New (2017 = 0)	0.0	0.1	0.0	0.4	1.2
Not exposed	Existing (2017 > 0)	9.7	9.3	35.0	44.3	6.5
Not exposed	New (2017 = 0)	0.0	5.4	0.0	25.8	7.5
Total		100.0	100.0	360.6	474.2	3.6

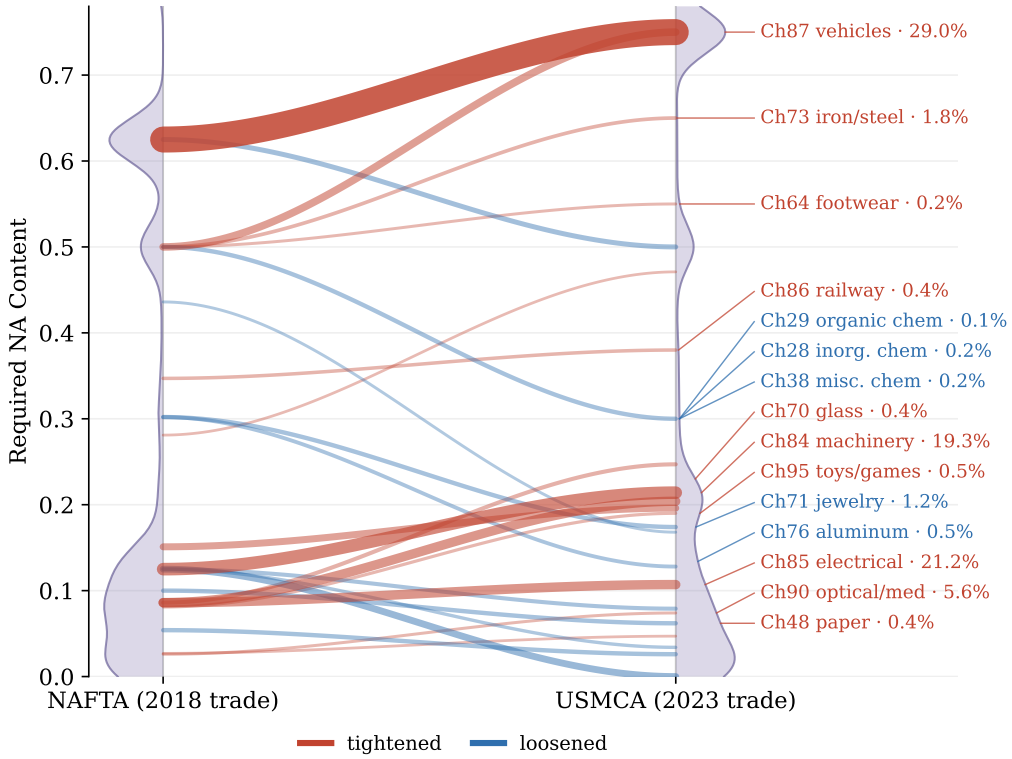
Notes: MX→US manufacturing trade (HS 16–97) in 2023 USD, from annual (2017) and monthly (2025) Comtrade data. Exposure to the 301 tariffs follows Bown (2021). NTR rates come from the USITC HTS and are trade-weighted within each cell, although in the (not exposed, new) cell only 0.6% of 2025 value carries a rate.

2.2 USMCA Preference Utilization

USMCA inherits the four-regime structure that NAFTA used for its Product Specific Rules of Origin, and adjusts each of these regimes asymmetrically. Under *Change in Tariff Classification (CTC)*, a good qualifies if its non-North American inputs undergo a specified tariff shift through production. *Regional Value Content (RVC)* requires that a minimum share of the good's value originate in North America. *Hybrid* regimes allow a good to qualify under

either CTC or RVC, whichever is easier to satisfy. Lastly, *Yarn-Forward* applies to textiles and requires that upstream yarn originate in North America. This last regime is omitted throughout the paper, given that its multi-stage production requirements go beyond the scope of the quantitative model. The largest change is in the RVC regime. The light-vehicle threshold rises from 62.5% to 75%, phased in over 2020–2023, and two requirements that did not exist under NAFTA are added to it, a Labor Value Content rule (40–45% of value produced by workers earning at least $16/hour) and a 70% North American steel and aluminum purchasing requirement. CTC tightens through HS-specific updates to the tariff shifts, which now tend to require a shift from a more granular subheading. Given that Hybrid rules are a combination of the two, they end up inheriting both of these changes.

Figure 2: The USMCA tightening of required North American content is asymmetric



Notes: Ribbons bundle HS6 subheadings moving identically NAFTA→USMCA; width proportional to average 2018–2023 MX→US export value. Violins are trade-weighted marginals (2018 for NAFTA, 2023 for USMCA); labels mark the fifteen HS2 chapters with the most trade in subheadings whose requirement moved, with each chapter's average 2018–2023 share of MX→US manufacturing trade.

Figure 2 puts every manufacturing subheading on one axis, which measures the North American content that it requires, and I draw each ribbon with a width that is proportional to the average MX→US exports of 2018–2023. In the case of an RVC subheading, its position is simply the threshold that the rule writes down. In the case of a CTC subheading, I compute the floor that the required tariff shift implies on the NA content of the final good, taking as

given the I-O input mix of the subheading. Hybrid subheadings prove to be harder to place, given that they carry both parameters. What I do is to put them on the axis of whichever parameter exists under both agreements (the RVC threshold whenever both of them carry one, and otherwise the CTC floor), and a subheading that is Hybrid under both agreements gets drawn on both axes. The violin on the left shows the trade-weighted distribution of required content under NAFTA and the one on the right the distribution under USMCA, whereas each ribbon follows a group of subheadings that moves together when going from one agreement to the other.

One can see that most of the mass of the USMCA distribution stays where the mass of NAFTA used to be, meaning that the tightening is concentrated in a group of products that come to form a second mode near 0.75, something that did not exist under NAFTA. Among the subheadings whose binding parameter is the RVC threshold (RVC and Hybrid), the share of MX→US trade with a required content of 70% or more goes from about 0% under NAFTA to about 30% under USMCA. Almost all of this upper mode is motor vehicles and parts (HS 87), 29% of MX→US manufacturing trade, which converge on the 75% threshold. Machinery (HS 84, 19% of trade) and electrical equipment (HS 85, 21%) account for most of the rest of the increase. About 12% of trade loosens (the blue ribbons).

Table 2 gives the direction of the change within each USMCA regime. In the RVC regime, 64.2% of 2023 trade is in an HS6 whose required content rose and 2.1% in one where it fell, and another 30% came over from CTC at the transition. In CTC about a quarter of the trade tightens, a quarter loosens and nearly half does not move. Hybrid is in between, with 37% tightening and 6% loosening. Weighting the three regimes by 2023 trade, about 40% of manufacturing trade faces a stricter requirement and 12% a looser one, 36% is unchanged and 10% switches between CTC and RVC.

Table 2: Direction of the NAFTA→USMCA rule change, by USMCA regime

USMCA regime	Within-regime direction of change (% of regime trade)			
	Tighter ($\Delta > 0$)	Unchanged ($\Delta = 0$)	Looser ($\Delta < 0$)	Regime switch (CTC↔RVC)
RVC	64.2	3.3	2.1	30.4
CTC	25.5	46.7	25.9	1.9
Hybrid	36.6	57.8	5.6	0.0
All three (aggregate)	40.5	36.1	11.8	10.0

Notes: Δ from Figure 2. The aggregate row is weighted by 2023 MX→US exports, and the 1.5% of trade that it does not cover is Yarn-Forward. Regime switches are crossings between pure CTC and pure RVC only (products moving into or out of Hybrid are in the tighter, unchanged and looser columns).

Table 3 reports how compliance moved in each of the three main regimes. From 2017 to 2024 the aggregate rate fell by about 8 percentage points, although not by the same amount

Table 3: USMCA compliance regimes: scope, trade share, and compliance change

	Trade share (%)				Compliance rate (%)				Avg NTR
Regime	2017	2024	2025 Jan–Feb	2025 Mar–Dec	2017	2024	2025 Jan–Feb	2025 Mar–Dec	(%)
RVC	31.6	31.4	28.3	27.5	94.5	80.8	78.2	84.4	7.6
CTC	37.3	35.9	36.3	33.0	30.8	31.6	30.3	84.3	2.6
Hybrid	29.5	31.9	34.5	38.7	46.4	34.2	29.1	85.9	1.4
Aggregate	98.4	99.1	99.1	99.2	55.9	48.0	43.6	85.0	3.7

Notes: The Feb/Mar cut isolates the IEEPA order (a 25% tariff on top of the NTR rate on Mexican imports outside USMCA preference, in force from March 4, 2025, with USMCA-compliant goods exempt from March 7). Compliance rates are value-weighted within each regime. Trade shares are computed over the full four-regime manufacturing universe, so what they leave out is the omitted Yarn-Forward regime.

everywhere, as how much a regime lost depends on whether its binding constraint is a value or a classification. The two regimes whose binding constraint depends on the *value* of inputs absorb almost all of the decline, as RVC falls 14 percentage points (from 94.5% to 80.8%) and Hybrid falls 12 percentage points (from 46.4% to 34.2%), whereas CTC barely moves. Given that the trade-weighted NTR on the three modeled regimes was 3.7% (Table 3),[3] the option value of skipping USMCA on eligible trade was small.

This changed in March 2025, when the IEEPA executive order imposed a 25% tariff on Mexican imports on top of the existing NTR rate while exempting goods meeting USMCA rules of origin, which raised the cost of failing to claim preference overnight. The response on the compliance margin shows to be immediate and large. In Jan–Feb 2025, aggregate value-weighted compliance was still below its 2024 level (∼44%), and over Mar–Dec 2025 it jumped to ∼85% (Table 3), where CTC and Hybrid nearly tripled and RVC went up by about six points. A jump of this size suggests to me that most of the gap observed before 2025 came from firms that preferred to source without any restrictions and to pay the NTR tariff, and that only started claiming preference once that tariff became large enough to matter.

These facts by themselves cannot tell us how costly compliance is for the firms that comply, how much of the decline before 2025 is due to USMCA and how much to the trade war, or what each of these policies would have done on its own. Answering these questions requires a model of how firms decide where to source from and whether to comply.

[3]Table 1 reports 3.6% because it averages over all MX→US manufacturing trade, including Yarn-Forward and products without a coded rule, weighted by 2017 and 2025 trade, whereas Table 3 covers only the three modeled regimes, weighted by 2017, 2024 and March–December 2025 trade.

3 Model

There are three things that these facts ask of the model. The first is that claiming preference has to be a decision of the firm, product by product, given that compliance moves within each regime and that not every shipment of the same product carries a claim. The second is that the RVC rule (a minimum North American share of the good's value) and the CTC rule (non-North American inputs must change tariff class in production) have to enter as separate technologies, since compliance moved differently across regimes in the same way that their rules differ. Lastly, the trade reorganization concentrates in 301-exposed products while Mexico bought more and more Chinese inputs, so firms need to choose their source countries, and where their inputs come from has to matter for whether they can comply. The model can then attribute the observed compliance changes to each policy, and say what compliance would have been under the trade war alone, under USMCA alone, under the higher tariff on non-compliant goods alone, and under all of them.

3.1 Environment

I develop a global sourcing model in the tradition of Eaton and Kortum (2002), Antràs et al. (2017), and Tintelnot (2017), which I extend with free trade agreement rules of origin and an endogenous compliance decision at the firm-product level. Firms in the model trade off cheaper inputs from global sourcing against the restrictions they must accept on their supply chain to sell tariff-free in the U.S. and Canada. They take the trade-policy parameters as given.

There is a set of source countries and a set of destination markets. Intermediates are produced in the country blocks

$$s \in \mathcal{S} = \{s_1, s_2, \ldots, s_n\},$$

of which the North American blocks (NA) count as originating under the agreement. Output is sold in the destination blocks

$$d \in \mathcal{D} = \{d_1, d_2, \ldots, d_m\},$$

which include the domestic market, Mexico (MX), as well as the U.S. and Canada (U.S. & CA), which is the only destination where the rules of origin apply. Note that the mechanisms of the model do not depend on the exact composition of $\mathcal{S}$ and $\mathcal{D}$. The blocks that are used in the estimation are listed in Section 4.

3.2 Consumers

Consumers in each destination d split total expenditure Y_d between manufactures and a homogeneous non-manufacturing good $C_{o,d}$ through a Cobb-Douglas upper tier,

$$U_d = C_{M,d}^{\chi_d} C_{o,d}^{1-\chi_d}, \qquad \chi_d \in (0,1), \tag{1}$$

where $C_{M,d}$ is the manufacturing composite and the non-manufacturing good is the numeraire ($p_o \equiv 1$). A constant share of expenditure falls on manufactures,

$$E_d \;\equiv\; \chi_d \, Y_d, \tag{2}$$

with the remainder $(1-\chi_d)Y_d$ spent on the non-manufacturing good. The manufacturing composite is itself a CES aggregate over a continuum of differentiated varieties $\omega \in \Omega_d$, with exogenous appeal weights $b_d(\omega) > 0$ scaling effective consumption:

$$C_{M,d} = \left(\int_{\omega \in \Omega_d} \left(b_d(\omega)\, q_d(\omega) \right)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \tag{3}$$

with $\sigma > 1$ the elasticity of substitution. Utility maximization subject to the manufacturing budget $\int p_d(\omega)\, q_d(\omega)\, d\omega = E_d$ yields the quantity demanded

$$q_d(\omega) \;=\; E_d \, P_d^{\sigma-1} \, p_d(\omega)^{-\sigma} \, b_d(\omega)^{\sigma-1}, \tag{4}$$

where $P_d = \left(\int \left(p_d(\omega)/b_d(\omega) \right)^{1-\sigma} d\omega \right)^{1/(1-\sigma)}$ is the CES price index of destination d. The only difference across destinations lies in what is endogenous. For the foreign markets $d \in \{\text{U.S. \& CA, RoW}\}$, total expenditure Y_d is exogenous, so E_d is taken as given and the share χ_d is absorbed into its calibrated level. For Mexico, on the other hand, manufacturing non-wage income Π_M is endogenous and non-manufacturing income Π_o is exogenous (both are defined in Section 3.4), so E_M moves with Mexican income $w_m \bar{L} + \Pi_M + \Pi_o$ through (35), which is what transmits the feedback from firm activity to domestic demand through Π_M.

3.3 Firms

Primitives. Each potential firm f in mass $\bar{M}$ is characterized by three objects: a product set $\mathcal{P}_f$ (empirically set at the HS6 level), a vector of country-specific sourcing efficiencies $\{a_{f,s}\}_{s \in \mathcal{S}}$, and a vector of destination-specific market reach $\{b_{f,d}\}_{d \in \mathcal{D}}$. The sourcing efficiency $a_{f,s}$ captures the firm's supplier networks and institutional relationships in source country s. For instance, U.S.-owned firms tend to have a high $a_{f,\text{U.S. \& CA}}$, Chinese-owned firms tend to

have a high $a_{f,CH}$, and so on. The market reach $b_{f,d}$ plays the analogous role on the demand side, scaling revenue at destination d in order to capture features such as market access, distribution networks, and brand recognition. Each product $p \in \mathcal{P}_f$ in turn determines the compliance regime and the NTR tariff that the firm faces.

Sourcing Strategy. Each firm sources a continuum of intermediate inputs $j \in [0,1]$ from source countries in $\mathcal{S}$. To source from country s, the firm knows how well it can source from it through $a_{f,s}$ and pays a fixed cost $w_m \kappa_s \geq 0$ in units of Mexican labor. Let $\mathcal{S}_f \subseteq \mathcal{S}$ denote the set of countries from which the firm sources. I refer to $\mathcal{S}_f$ as the firm's *sourcing strategy*, and assume that it is shared across all of the firm's products and destinations.[4]

Conditional on paying the fixed cost, the firm draws input-level productivities $z_s(j) \sim$ Fréchet$(T_s a_{f,s}^{\theta}, \theta)$ independently across inputs and source countries. These draws capture every firm-, input- and country-specific factor that plays a role in the realized landed cost, such as the quality of the supplier match, the reliability of shipping or how well the product fits. $T_s > 0$ is the absolute sourcing advantage of country s, θ governs the dispersion of the productivity draws, and the firm-country sourcing efficiency $a_{f,s}$ shifts the location of the distribution upward, which raises the likelihood that country s offers a lower landed cost for a given input. The per-input landed cost is

$$c_{f,s}(j) = \frac{c_s \tau_s}{z_s(j)} \tag{5}$$

where c_s is the cost of inputs in source country s and τ_s the iceberg trade cost of sourcing from s. The firm then picks, for each input j, the source that offers it at the lowest cost, $s^*(j) = \arg\min_{s \in \mathcal{S}_f} c_{f,s}(j)$. The intermediate cost index comes from integrating these per-input minima over the continuum of inputs. Under the Fréchet distribution it has the following closed form:

$$\begin{aligned} c_f^M(\mathcal{S}_f) &= \int_0^1 \min_{s \in \mathcal{S}_f} \frac{c_s \tau_s}{z_s(j)} \, dj \\ &= \Gamma\left(1 - \frac{1}{\theta}\right) \left(\sum_{s \in \mathcal{S}_f} T_s a_{f,s}^{\theta} (c_s \tau_s)^{-\theta} \right)^{-1/\theta} . \end{aligned} \tag{6}$$

The expected share of intermediates sourced from country s is the integral over j of the indicator that s offers the minimum landed cost, which the same algebra evaluates as

[4] Customs data record input origin and output destination separately, but not which input made which output, so a shared $\mathcal{S}_f$ is what the data support.

$$\mu_{f,s}(\mathcal{S}_f) = \int_0^1 \mathbf{1}\left\{ \frac{c_s \tau_s}{z_s(j)} = \min_{s' \in \mathcal{S}_f} \frac{c_{s'} \tau_{s'}}{z_{s'}(j)} \right\} dj$$
$$= \frac{T_s a_{f,s}^{\theta} (c_s \tau_s)^{-\theta}}{\sum_{s' \in \mathcal{S}_f} T_{s'} a_{f,s'}^{\theta} (c_{s'} \tau_{s'})^{-\theta}}. \tag{7}$$

Production Technology. Each active firm produces output by combining Mexican labor ℓ_f with an intermediate composite M_f through a Cobb-Douglas technology:

$$q_f = \ell_f^{\gamma} M_f^{1-\gamma}, \tag{8}$$

where $\gamma \in (0,1)$ is the value-added share of labor. Labor is hired locally at the wage w_m, and by construction its cost counts as North American value added. The intermediate composite M_f in turn aggregates the continuum of inputs $j \in [0,1]$, where each input is sourced from the country in the firm's sourcing set that offers it at the lowest cost. Note how this allows the possibility for a firm to comply with the trade agreement requirements not only by sourcing inputs from North America, but also through the value added by Mexican labor.

Combining (8) with (6), cost-minimization over labor and the intermediate composite yields the Cobb-Douglas unit cost:

$$c_f(\mathcal{S}_f) = B\, w_m^{\gamma} \left[c_f^M(\mathcal{S}_f) \right]^{1-\gamma}, \tag{9}$$

where $B \equiv \gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)}$ is a constant derived from the cost-minimization problem.

Compliance: Three Regimes.[5] Rules of origin bind only on U.S. & CA-bound output. For each product $p \in \mathcal{P}_f$, the firm chooses a compliance indicator $d_{fp} \in \{0,1\}$, where $d_{fp} = 1$ means that the firm accepts the applicable input restriction in exchange for tariff-free access to the U.S. & CA market, whereas $d_{fp} = 0$ means that it pays the NTR tariff τ_p^{NTR} on U.S. & CA-bound shipments and sources freely. The different regimes are quantified as follows:

- **Change in Tariff Classification (CTC).** A CTC rule for output p names, at a treaty-determined HS depth (chapter, heading or subheading), a specific set $\mathcal{P}(p)$ of input HS classes whose tariff classification must change *through production* in order for the input to qualify. A non-NA input from $\mathcal{P}(p)$ whose tariff class is unchanged through production fails the rule, whereas an NA-sourced input always passes, regardless of its classification. I take the I-O technology at output p as binding, meaning that any input

[5] Yarn-Forward is omitted given that textiles and apparel account for under 2% of MX→US manufacturing trade, and that their multi-tier yarn→fabric→garment origin chain goes beyond the model's single-tier sourcing.

in $\mathcal{P}(p)$ to which the I-O table assigns a positive expenditure share at p is interpreted as failing the classification-change test under p's production technology, and must therefore be NA-sourced for compliance. The rule then reduces to a single product-specific scalar, where $\phi_p(j)$ is the I-O expenditure share of input class j in p's intermediate bundle:

$$\mu_p^{CTC} \;=\; \sum_{j \in \mathcal{P}(p)} \phi_p(j),$$

which is the I-O share of p's intermediate bundle allocated to those restricted-and-unchanged classes. Through the lens of the model, the rule maps to a subset of the firm's continuum of input varieties $j \in [0,1]$ of measure μ_p^{CTC} that must be NA-sourced, while the residual measure $1-\mu_p^{CTC}$ is drawn from the unrestricted set $\mathcal{S}_f$. Given that input varieties are drawn iid Fréchet across countries, which varieties end up in the restricted subset does not matter for its cost, only its measure does, meaning that the compliant intermediate cost takes the form of a Cobb-Douglas aggregation between the restricted (NA-only) and the unrestricted sub-continua:

$$c_{f,p}^{M,CTC}(\mathcal{S}_f) = \Gamma\big(1-\tfrac{1}{\theta}\big)\left(\sum_{s \in NA \cap \mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}\right)^{-\mu_p^{CTC}/\theta}\left(\sum_{s \in \mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}\right)^{-(1-\mu_p^{CTC})/\theta}. \tag{10}$$

Both trade agreements further allow a *de minimis* relaxation, under which CTC is treated as satisfied whenever non-conforming non-originating inputs account for at most $\delta_p = 10\%$ of the good's value under USMCA and 7% under NAFTA. Given that μ_p^{CTC} is a share of intermediate spending and that intermediates make up a fraction $1-\gamma$ of the good's cost, the tolerance trims the binding restricted measure to $\max\big(\mu_p^{CTC} - \delta_p/(1-\gamma),\, 0\big)$, where the good's value is measured at total cost, which is the same basis as the net-cost RVC thresholds below. This effectively allows firms to have a small margin of deviation from the CTC requirement without losing preference eligibility.

- **Regional Value Content (RVC).** The North American content share of production must reach a product-specific threshold μ_p^{RVC}. Given that Mexican labor counts as North American content, the content share under an unrestricted sourcing strategy is

$$\gamma + (1-\gamma) \sum_{s \in NA \cap \mathcal{S}_f} \mu_{f,s}(\mathcal{S}_f) \geq \mu_p^{RVC}, \tag{11}$$

where the first term is the automatic contribution of Mexican labor and the second is the NA share of the firm's intermediate sourcing. The rule is met when the value added

both from labor and North American manterial exceeds the threshold determined by the free trade agreement.

- **Hybrid.** The firm may qualify under either rule and takes the cheaper compliant cost:

$$c_{f,p}^{Hyb}(\mathcal{S}_f) \;=\; \min\!\big(c_{f,p}^{CTC}(\mathcal{S}_f),\; c_{f,p}^{RVC}(\mathcal{S}_f)\big)\,.$$

Note that the firm decides on compliance separately for each product and destination (only U.S. & CA-bound shipments face the rules), while its sourcing strategy $\mathcal{S}_f$ is the same for all of its products and destinations. A firm can therefore comply for one product and not for another, choosing its compliance vector $\{d_{fp}\}$ product by product, but once it has paid the sourcing fixed costs it buys from the same countries for every product. I keep a single sourcing strategy per firm for two reasons. It keeps the discrete-choice problem tractable, and it is what the customs data allow, as they record the import composition of the firm but not which inputs go into which product.

Conditional on $\mathcal{S}_f$ and the per-product compliance indicator d_{fp}, the destination-specific landed unit cost is

$$c_{f,p,d}(\mathcal{S}_f, d_{fp}) \;=\; \begin{cases} c_{f,p}(\mathcal{S}_f) & \text{if } d = \text{U.S. \& CA and } d_{fp} = 1, \\ c_f(\mathcal{S}_f)\,(1+\tau_p^{NTR})^{\sigma/(\sigma-1)} & \text{if } d = \text{U.S. \& CA and } d_{fp} = 0, \\ c_f(\mathcal{S}_f) & \text{if } d \in \{\text{MX, RoW}\}, \end{cases} \tag{12}$$

where $c_f(\mathcal{S}_f)$ is the unrestricted unit cost from (9) and $c_{f,p}(\mathcal{S}_f)$ the compliant unit cost, which is $c_{f,p}^{CTC}(\mathcal{S}_f)$, $c_{f,p}^{RVC}(\mathcal{S}_f)$ or $c_{f,p}^{Hyb}(\mathcal{S}_f)$ depending on the regime that governs product p. Note that the choice $d_{fp} = 1$ requires at least one North American country in $\mathcal{S}_f$ under every regime, given that both the CTC restriction μ_p^{CTC} and the RVC restriction $\alpha_{f,p}^*$ draw the restricted inputs from NA sources. Specifically, a firm that complies produces at the restricted cost $c_{f,p}$ and does not pay a tariff, whereas a firm that does not comply produces at the unrestricted cost c_f but pays the NTR tariff on its U.S. & CA-bound revenue. The exponent $\sigma/(\sigma-1)$ has to do with the duty being ad valorem on the fob value, meaning that this is the exponent that turns it into a wedge on cost. The U.S. & CA buyer pays $p(1+\tau_p^{NTR})$ and the firm receives $p = \frac{\sigma}{\sigma-1}c_f$, while the duty $\tau_p^{NTR}\,p\,q$ goes to the U.S. Treasury. The fob revenue and profit of a non-compliant product then scale as $(1+\tau_p^{NTR})^{-\sigma}$, which is exactly what the wedge gives in (13). MX and RoW shipments are unaffected by the compliance choice.

Destination Markets. Each firm chooses a destination set $\mathcal{D}_f \subseteq \mathcal{D}$, and pays a fixed entry cost $w_m\kappa_d$ in units of Mexican labor for each $d \in \mathcal{D}_f$. Each product of an active firm

is a differentiated variety at every destination that it serves. Given constant-markup pricing over the duty-free unit cost, the firm's fob revenue from product p at destination d is

$$R_{f,p,d}(\mathcal{S}_f, b_{f,d}) \;=\; \varpi_{f,p}\, E_d\, P_d^{\sigma-1} \left(p_{f,p,d}(\mathcal{S}_f, d_{fp})\right)^{1-\sigma} b_{f,d}^{\sigma-1}, \tag{13}$$

where $p_{f,p,d}(\mathcal{S}_f, d_{fp}) = \frac{\sigma}{\sigma-1}\, c_{f,p,d}(\mathcal{S}_f, d_{fp})$ is the constant-markup price over the destination-specific landed unit cost from (12), and $\varpi_{f,p}$ is product p's share of the firm's observed exports times its number of products, so that the weights average to one within the firm.

Profits and the Firm Problem. The firm jointly chooses its sourcing set $\mathcal{S}_f$, its destination set $\mathcal{D}_f$ and a per-product compliance vector $\{d_{fp}\}_{p\in\mathcal{P}_f}$ in order to maximize expected profits:

$$\begin{aligned}
\pi_f = \max_{\mathcal{S}_f,\, \mathcal{D}_f,\, \{d_{fp}\}} \; & \frac{1}{\sigma} \sum_{d\in\mathcal{D}_f} \sum_{p\in\mathcal{P}_f} R_{f,p,d} \;-\; w_m \sum_{s\in\mathcal{S}_f} \kappa_s \;-\; w_m \sum_{d\in\mathcal{D}_f} \kappa_d \\
\text{subject to } \; & \mathcal{S}_f \subseteq \mathcal{S}, \quad \mathcal{D}_f \subseteq \mathcal{D}, \quad d_{fp} \in \{0,1\}, \\
& d_{fp} = 1 \implies \begin{cases} \mu_p^{RVC} \geq \mu_p^{CTC} & p \in \text{CTC}, \\ \gamma + (1-\gamma)\mu_f^{NA}(\mathcal{S}_f) \geq \mu_p^{RVC} & p \in \text{RVC}, \\ \text{either condition} & p \in \text{Hybrid} \end{cases}
\end{aligned} \tag{14}$$

where $\mu_f^{NA}(\mathcal{S}_f) \equiv \sum_{s\in NA\cap\mathcal{S}_f} \mu_{f,s}(\mathcal{S}_f)$ is the North American share of the firm's intermediate bundle. Since every product and destination of the firm draws on the same sourcing set, the firm has only one such share. It chooses its North American content for its input pool as a whole, and not product by product. A Hybrid product qualifies under either rule.

As in Antràs et al. (2017), solving (14) is a combinatorial problem, in which the firm evaluates every non-empty sourcing set against every destination set and selects the pair that yields the highest profit, with the additional layer that, for each pair, it decides product by product whether to comply (Appendix A.5).

Endogenous Entry and Exit. Let $\pi_f(w_m)$ denote the maximized expected profit from (14), with the dependence on the Mexican wage made explicit. Firm f is *active* if and only if some non-empty $(\mathcal{S}_f, \mathcal{D}_f)$ yields positive profits:

$$f \in \mathcal{A}(w_m) \quad \Longleftrightarrow \quad \pi_f(w_m) > 0. \tag{15}$$

The endogenous active set $\mathcal{A}(w_m) \subseteq [0, \bar{M}]$ is the subset of potential firms for whom some sourcing, destination and compliance strategy is profitable. Inactive firms do not produce, source inputs, or employ Mexican labor. As in Tintelnot (2017), the mass of potential firms

$\bar{M}$ is taken as a primitive, so that there is no free-entry condition. Considering that firm heterogeneity is multidimensional (the full sourcing-efficiency vector $\{a_{f,s}\}$ and market-reach vector $\{b_{f,d}\}$), there is no single productivity cutoff separating active from inactive firms.

3.4 Market Clearing and Equilibrium

Mexico is endowed with a fixed labor force L, which is mobile across the manufacturing sector modeled above and a homogeneous non-manufacturing sector. The non-manufacturing sector is perfectly competitive and produces the numeraire good with labor alone under constant returns, $Q_o = A_o L_o$, where A_o is its labor productivity. With free entry, price equals marginal cost, $p_o = w_m/A_o$, and since the non-manufacturing good is the numeraire ($p_o \equiv 1$), the Mexican wage is tied to productivity in that sector,

$$w_m = A_o. \tag{16}$$

Manufacturing therefore hires at a wage that it is not able to move, given that workers can always earn w_m in an outside sector which is competitive and large enough to absorb them, so that labor moves between the two sectors without any change in the wage. Antràs et al. (2017) close their model in the same way.[6]

The labor force L then divides itself between the two sectors. Manufacturing firms hire as many workers as they want at $w_m = A_o$, and the rest, $L_o = L - L_m \geq 0$, work in the non-manufacturing sector. Letting $VC_f \equiv \frac{\sigma-1}{\sigma}\sum_{d,p} R_{f,p,d}$ denote total variable cost, manufacturing labor demand aggregates over the active set $\mathcal{A} \equiv \mathcal{A}(w_m)$:

$$L_m = \int_{f\in\mathcal{A}} \Big[\big(\gamma + (1-\gamma)\mu_{f,MX}\big)\tfrac{VC_f}{w_m} \;+\; \sum_{s\in\mathcal{S}_f} \kappa_s \;+\; \sum_{d\in\mathcal{D}_f} \kappa_d \Big] df, \tag{17}$$

where the bracket adds up production labor (the γ share of variable cost under Cobb-Douglas), Mexican-sourced intermediates (the $(1-\gamma)\mu_{f,MX}$ share) and the fixed costs of sourcing and destination entry. Note that no wage is solved for through equation (17). I use it as an accounting identity that reads L_m off the firm-level solution, and the pinned-wage closure is interior as long as $L_m \leq L$.

For the foreign destinations $d \in \{\text{U.S. \& CA, RoW}\}$, manufacturing expenditure and prices (E_d, P_d) are taken as exogenous, considering that Mexico is a small open economy relative to those markets and its firms' decisions are not expected to materially move external incomes or price levels. The Mexican pair (E_M, P_M), on the other hand, is determined within

[6] Non-manufacturing accounts for about 75% of Mexican employment (INEGI, ENOE), which supports the perfectly-elastic labor-supply pole.

the model, as described in Appendix A.8.

The Mexican price index P_M in turn aggregates the prices of Mexican-firm varieties sold at MX through the CES dual of (3):

$$\left(P_M\right)^{1-\sigma} = \int_{f\in\mathcal{A}:\, MX\in\mathcal{D}_f} \sum_{p\in\mathcal{P}_f} \left(\frac{p_{f,p,MX}}{b_{f,MX}}\right)^{1-\sigma} df. \tag{18}$$

The welfare metric that I report in the counterfactuals is Mexican consumer real income under the Cobb-Douglas upper tier. The ideal price index dual to $U = C_M^{\chi} C_o^{1-\chi}$ is the geometric mean $P \propto P_M^{\chi}\, P_o^{1-\chi}$, so deflating aggregate nominal Mexican income by P gives

$$W_M \;=\; \frac{w_m L \;+\; \Pi_M + \Pi_o}{P_M^{\chi}\, P_o^{1-\chi}}, \tag{19}$$

where the three numerator components are, in order, the economy-wide wage bill (both sectors pay $w_m = A_o$ over the fixed labor supply L), manufacturing non-wage income

$$\Pi_M \;=\; \int_{f\in\mathcal{A}} \pi_f \, df, \tag{20}$$

and non-manufacturing income Π_o. Given that this sector is perfectly competitive and operates under constant returns, free entry drives economic profits to zero, so that $\Pi_o \equiv 0$. Taking $P_o = 1$ as the numeraire, the denominator of (19) reduces to P_M^{χ}. Given that $w_m = A_o$ and L do not move under the pinned-wage closure, any movement in the numerator of the welfare measure has to come through Π_M.

Putting all of the above together, I can now define the equilibrium of the model.

Given preference $(\sigma,\, \chi)$, technology $(\theta,\, \gamma,\, A_o)$, firm $(\mathcal{P}_f,\, \{a_{f,s}\},\, \{b_{f,d}\})$, policy $(\{\mu_p^{CTC},\, \mu_p^{RVC}, \tau_p^{NTR}\})$, foreign-market $(\{E_d, P_d\}_{d\neq MX},\, \{T_s, c_s, \tau_s\})$, and fixed-cost $(\{\kappa_s, \kappa_d\})$ primitives, together with the Mexican labor force L, an equilibrium is a tuple

$$\left(\mathcal{A},\; \{\mathcal{S}_f, \mathcal{D}_f, (d_{fp})_{p\in\mathcal{P}_f}\}_{f\in\mathcal{A}}\right)$$

such that, at the pinned wage $w_m = A_o$:

(i) Consumers satisfy the Cobb-Douglas upper tier (1) and CES demand (4) at each destination.

(ii) Each active firm $f \in \mathcal{A}$ solves (14) for $(\mathcal{S}_f, \mathcal{D}_f, \{d_{fp}\})$, taking the domestic pair (E_M, P_M) and the foreign pairs $\{(E_d, P_d)\}_{d\neq MX}$ as given.

(iii) Entry satisfies (15): $\mathcal{A} = \{f : \pi_f(w_m) > 0\}$.

(iv) Domestic demand is consistent: E_M satisfies (35) with Π_M from (20), and P_M satisfies (18), both at the firms' choices.

Given that the wage is pinned, solving for the equilibrium only requires iterating on the domestic pair (E_M, P_M) until condition (iv) holds, which I describe in Appendix A.8.

4 Calibration

In the calibration, the sourcing side is divided into six country blocks,

$$\mathcal{S} = \{\text{U.S. \& Canada (U.S. \& CA), Mexico (MX), Europe (EU),}\\ \text{China (CH), Asia excl. China (AX), Rest of World (RoW)}\},$$

and the sales side into three destination blocks,

$$\mathcal{D} = \{\text{U.S. \& Canada (U.S. \& CA), Mexico (MX), Rest of World (RoW)}\}.$$

I pool the U.S. and Canada on both the sourcing and the demand side. The two share a compliance regime, and from Mexico's perspective they play the same role in the model, that of a single tariff-free North American partner. China is kept apart from the rest of Asia in order to be able to tariff a non-FTA partner on its own. Having separate destinations in turn allows the firm to decide on compliance product by product according to the share of its revenue that goes to the U.S. & CA, instead of taking an all-or-nothing decision at the plant level.

4.1 Data

I take the model to firm-level Mexican data drawn from two sources: the *Profile of Export Manufacturing Enterprises* (PEME) customs microdata and the Mexican *Economic Census* (CE). PEME records the universe of Mexican customs transactions at the firm $\times$ HS8 $\times$ partner $\times$ year level, and from it I construct firm-level import-source shares $\mu_{f,s}$ for $s \neq \text{MX}$ and export-destination shares $\omega_{f,d}$ for $d \in \{\text{U.S. \& CA, RoW}\}$. From the CE, which covers the population of Mexican manufacturing establishments in 2019 and 2024, I recover each firm's domestic material purchases, domestic sales, employment and paid wage bill, ownership and industry. The CE is collected at the establishment level, so I add up the establishments of each firm using the INEGI firm identifier and I take the industry and ownership of the firm from its largest establishment. Given that PEME uses this same identifier, linking the two datasets is straightforward, and for every matched firm I end up observing its census

characteristics alongside its customs sourcing and destination shares. Each census asks about the previous year, which is why the 2019 and 2024 censuses are matched to PEME 2018 and 2023. Every peso magnitude of the 2018 period is expressed in constant 2023 pesos. The matches are described in Appendix C.

The firms that I model individually are the census firms that match a PEME firm and that trade on at least one side, meaning that they either sell abroad or import materials. This includes the import-only firms, that is, those that buy abroad but export nothing. Each of these firms gets one potential export product, the modal HS6 of its six-digit SCIAN class in the Pierce and Schott (2012) concordance, weighted by Mexican exports to the U.S. and Canada. Should its imputed market reach take it into the U.S. & CA market in some counterfactual, the firm faces the rule and the NTR rate of that product. The rest of the census firms, those without a customs record and those with a record but no trade, I collapse into bins by NAICS-3 industry. These firms are of all sizes and their share of manufacturing activity is not small, so they still matter for the Mexican price index. They enter it through an additive term K in the CES sum, $P_M^{1-\sigma} = T + K$, where T aggregates the price terms of the traders and K is calibrated to match the baseline share of the non-traders. I model the non-traders as firms that source and sell only at home at the pinned wage. Their prices therefore do not move in the counterfactuals, and neither does K, although K does dampen the response of P_M since the traders are only a part of the CES sum. The sales of the non-traders do move with Mexican demand, and their variable profits (a fraction $1/\sigma$ of those sales) are part of manufacturing non-wage income Π_M.

The firm-level block runs on the traders, 5,726 firms in 2018 and 5,148 in 2023. Table 4 lists the parameters, which I calibrate in two layers. Some are fixed outside of the model, either from data or from the literature. The rest are structural, and I estimate them through model inversions, Heckman-corrected regressions and a moment-inversion estimator that matches aggregate participation shares.

4.2 Externally Calibrated Parameters

Preferences (σ). Following Antràs et al. (2017), I pin the CES elasticity σ from the firm's first-order condition under monopolistic competition, under which revenue equals a constant markup over variable cost, $R_f = \frac{\sigma}{\sigma-1} VC_f$, with $VC_f = wL_f + M_{t,f}$. I take the median of the firm-level markup R_f/VC_f across the CE manufacturing firms with a positive wage bill and positive materials in each census wave, after trimming the markups below 1 and above 10 as in Antràs et al. (2017), pool the two medians with weights proportional to each wave's total revenue, and invert the pooled markup, so that the same σ enters both periods. Pooling is a deliberate choice, as it imposes that σ, a deep preference parameter, is constant over the

Table 4: Calibration: Parameter Values and Identification

Parameter	**Object**	**Identified from**
A priori / data		
σ	CES elasticity	Unweighted median firm markup $R_f/VC_f = \sigma/(\sigma-1)$ (CE)
γ_s	Labor share	Sector (NAICS-3)cost share $\sum wL / \sum(wL + M_t)$ (CE)
$\mu_p^{CTC},\ \mu_p^{RVC}$	Product RoO parameters	NAFTA/USMCA + INEGI I-O
τ_p^{NTR}	NTR tariff	USITC
$E_d,\ P_d,\ d \in$ {U.S. & CA, RoW}	Foreign mfg. expenditure / price	World Bank WDI
$\psi_s \equiv T_s(c_s\tau_s)^{-\theta}$	Country sourcing shifter	Mean firm sourcing shares (CE and PEME)
L	Mexican labor force	ENOE economically active population
A_o	Non-mfg. productivity	Annual wage per worker (CE)
$\theta = 1.79$	Fréchet shape	Antràs et al. (2017)
Structural		
$a_{f,s}$ (per source country)	Firms' sourcing efficiency	Model based inversion and Heckman two-step over missing cells
$b_{f,d}$ (per destination)	Firms' market reach	Model based inversion and Heckman two-step over missing cells
κ_s (per source)	Source-entry fixed cost	Moment inversion on source participation shares
κ_d (per destination)	Destination-entry fixed cost	Moment inversion on destination participation shares
Equilibrium		
L_m	Manufacturing employment	Labor allocation (17) (non-mfg. residual)
$E_M,\ \Pi_M,\ P_M$	Equilibrium MX expenditure / income / price	Domestic-demand fixed point (35), (18) and (20)

five-year window, meaning that the decomposition in Section 6 is attributable to the inputs that do vary. The median markup is 1.64 in the 2019 wave and 1.47 in the 2024 wave, and the pooled markup is 1.54, which inverts to $\sigma = 2.84$.

Technology (γ). Under Cobb-Douglas production, the labor share of variable cost equals the production-function exponent (Appendix A.2), which justifies calibrating γ as a cost share. Given that input intensity differs sharply across manufacturing branches, I compute the share at the NAICS-3 sector level as a ratio of sums, $\gamma_s = \sum wL / \sum(wL + M_t)$, where the paid wage bill wL and materials purchases M_t are aggregated across *all* CE manufacturing firms in sector s (and not only across the traders of the model block), and then assign each model firm its sector's value. The sector shares are computed within each census wave and then frozen at their simple mean across the two waves. Appendix Figure 8 shows the per-wave and pooled values across the 21 NAICS-3 sectors. When weighted by the wage bill of the estimation panels, the pooled shares aggregate to 0.203 (2018) and 0.206 (2023).

Dispersion (θ). The Fréchet shape governing the dispersion of firm-level sourcing efficiency is set to $\theta = 1.79$ following Antràs et al. (2017).

Foreign-market block: country sourcing shifters (ψ_s). The bundle $\psi_s \equiv T_s(c_s\tau_s)^{-\theta}$ is identified from the mean sourcing shares of the firms. Given that T_s, c_s and τ_s cannot be identified separately, I work with ψ_s as a whole and normalize $\psi_{\text{U.S. \& CA}} \equiv 1$. The reason behind taking U.S. & CA as the normalizing coordinate is that it is the non-home block with the highest participation, which in turn provides a stable non-zero denominator for the firm-level ratio $\mu_{f,s}/\mu_{f,\text{U.S. \& CA}}$, which the structural step below relies on. Setting $a_{f,s} \equiv 1$ at this stage, which is without loss of generality, reduces (7) to $\bar{\mu}_s/\bar{\mu}_{\text{U.S. \& CA}} = \psi_s$, and each remaining $\hat{\psi}_s$ is then simply a ratio of sample means of the CE–PEME sourcing shares. I follow by restoring the firm-specific $a_{f,s}$ in Section 4.3.

Foreign-market block: manufacturing expenditure and prices (E_d, P_d). The CES revenue equation $R_{f,p,d} = \varpi_{f,p} E_d P_d^{\sigma-1} p_{f,p,d}^{1-\sigma} b_{f,d}^{\sigma-1}$ takes the manufacturing expenditure E_d and the price P_d as exogenous on the foreign side, with the upper-tier share χ_d absorbed into E_d (2). I normalize $E_d \equiv P_d \equiv 1$ in 2018 for both $d \in \{\text{U.S. \& CA, RoW}\}$. For 2023, I take the 2018→2023 growth of nominal GDP and of the implicit GDP deflator from the World Bank WDI and set E_d to their ratio. The countries are aggregated following the three destinations of the model, with U.S. & CA = USA + CAN and RoW = WLD − USA − CAN − MEX. Unlike the foreign pairs, the Mexican pair is not anchored to any data. What I do in the baseline is to take (Π_M, P_M) from the master solve, through the aggregation identities (18) and (20) at the pinned wage $w_m = A_o$. In the counterfactual exercises of Section 6, on the other hand, (E_M, P_M) moves together with income through (35).

Policy (μ_p^{CTC}, μ_p^{RVC}, τ_p^{NTR}). The product-level RoO parameters $(\mu_p^{CTC}, \mu_p^{RVC})$ come

from USMCA Annex 4-B[7] and from Conconi et al.'s (2018) coding of NAFTA Annex 401, and are refined with the INEGI 2018 input-output table in order to capture each product's input intensity. INEGI's I-O table is available at a six-digit SCIAN disaggregation, so I used Pierce and Schott (2012)'s HS-NAICS concordance tables to match the specific input content requirements determined at the HS system to the NAICS input requirements. More information on the matching between these two classifications is available in Appendix B. NTR tariffs τ_p^{NTR} come from the USITC 2023 Harmonized Tariff Schedule. RVC thresholds enter in their *net-cost* formulation rather than in their *transaction value* formulation (which is roughly 10pp higher), given that the condition (11) is a cost share. I attach every parameter at the HS6 level. Annex 4-B writes its product-specific rules at that level, and it is also where CTC, RVC and Hybrid can be told apart without ambiguity. Since customs codes change across HS revisions, the parameter file keeps codes from HS2012, HS2017 and HS2022, linked through the UN Stats conversion tables, so a code from either year still finds its rule. Coverage is 98% of Mexican manufacturing exports to the U.S. in 2018 and again in 2023, and what is left out is Yarn-Forward.

One detail matters when computing μ_p^{CTC} from the INEGI input-output table, and it is the HS depth at which the rule is written. I condition on it, so that the HS6 parameter reflects the restriction that actually binds. Laptops are a good example of why. For HS 8471.30 (portable automatic data processing machines), USMCA Annex 4-B specifies the following CTC restriction:

> *A change to a good of subheading 8471.30 from any other good within that subheading or any other subheading, except from subheading 8471.41 through 8471.50.*

What the carve-out does is close a substantial-assembly loophole. Think of a Mexican plant that imports a processing-unit module from China under HS 8471.50, already almost finished, and puts it into a chassis. Without the exception that laptop would meet the tariff shift, simply because 8471.50 and 8471.30 are different subheadings. In the input-output table, inputs in the 8471.41–50 range are 14.6% of the intermediate spending of HS 8471.30, and this is what gives $\mu^{CTC}_{8471.30} = 0.146$.[8]

[7] Given that Annex 4-B is only available as legal text, with no machine-readable version of it, I wrote a parser that goes through the annex and records every product-specific rule (its CTC clauses, RVC thresholds and de minimis exceptions) in an HS6-level dataset. This turns out to be feasible because the treaty relies on a small number of phrases that it repeats throughout ("a change to X from any other Y", "a regional value content of not less than Z percent"), so what the parser does is to match each rule to one of these templates and read the regime and the thresholds off of it. Whenever a rule is written for a whole chapter or heading, it is passed down to all of its HS6 subheadings, unless a subheading has a rule of its own.

[8] In the parameter file that the estimation uses, this rule is coded as a change of subheading. The clause "any other good within that subheading" exempts the inputs classified in 8471.30 itself, but this coding restricts them as well, which adds another 0.3% of spending and gives $\mu^{CTC}_{8471.30} = 0.149$.

Table 5 summarizes the resulting HS6 parameter universe that is used as a model input, and Figure 3 below shows the underlying trade-weighted distributions of μ_p^{CTC} and μ_p^{RVC}. On average across HS6 subheadings, pure-RVC coverage collapses to an automotive core at the 75% net-cost threshold, Hybrid coverage roughly doubles, and the CTC input restriction μ_p^{CTC} tightens on both its pure and its Hybrid arm.

Table 5: NAFTA vs USMCA Policy Parameters: Summary Stats

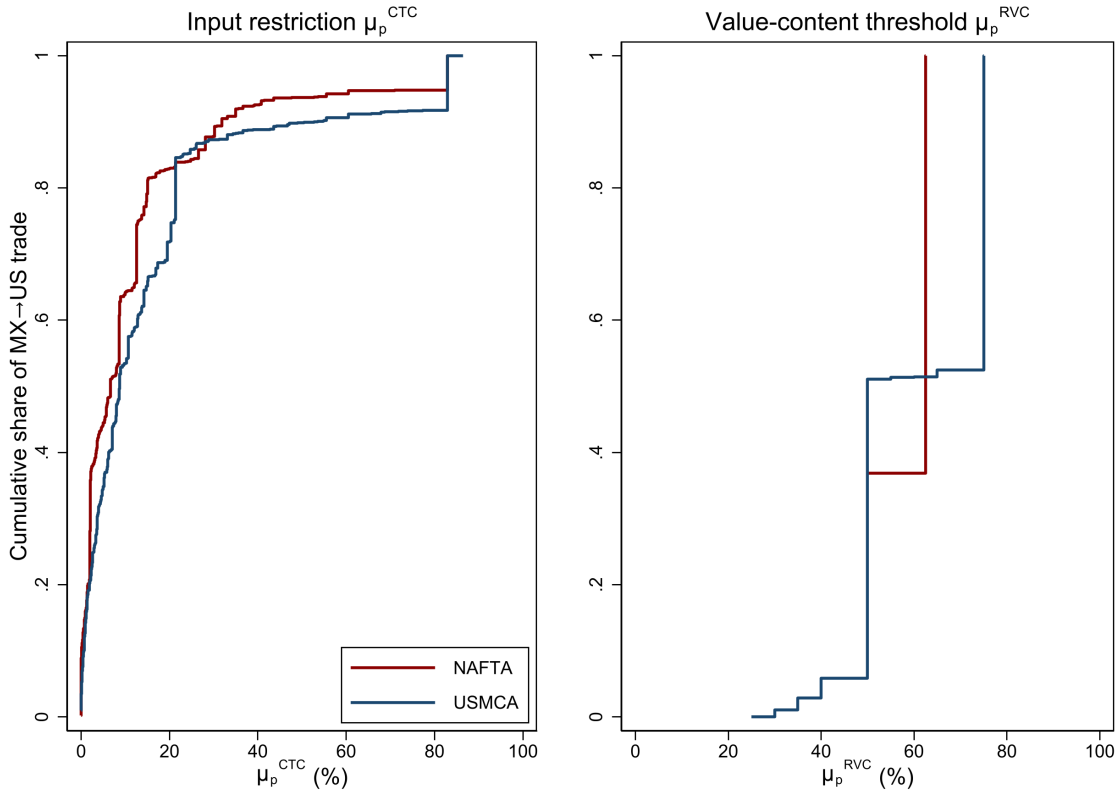
	NAFTA	**USMCA**	**Δ**
CTC regime			
HS6 subheadings	3,284	3,146	−138
μ_p^{CTC} (mean / median)	14.4 / 4.9	19.0 / 7.7	+4.6 / +2.8
RVC regime			
HS6 subheadings	717	79	−638
μ_p^{RVC} (mean / median)	50.0 / 50.0	71.9 / 75.0	+21.9 / +25.0
Hybrid regime			
HS6 subheadings	818	1,575	+757
μ_p^{CTC} (CTC arm, mean / median)	4.5 / 0.4	23.6 / 16.3	+19.1 / +15.9
μ_p^{RVC} (RVC arm, mean / median)	50.7 / 50.0	40.9 / 50.0	−9.8 / 0.0
NTR tariff τ_p^{NTR} (common across policy regimes)			
		Equal-wtd	*TW (2023)*
τ_p^{NTR} (mean / median)		3.3 / 2.4	3.3 / 1.4
Share with $\tau_p^{NTR} > 0$		58.8%	61.5%
Share with $\tau_p^{NTR} \geq 5$		24.2%	13.9%

Notes: Values in %; Δ in pp. Equal-weighted across the 5,077 HS6 subheadings of the parameter file in chapters 16–97, which include codes of the HS2012, HS2017 and HS2022 revisions. TW (2023): weighted by 2023 U.S. imports from Mexico (UN Comtrade). RVC on net-cost basis.

Closure ($\bar{L}$, A_o, χ). The labor-market closure adds a competitive non-manufacturing sector that pins the Mexican wage. Its productivity A_o sets the level of that wage and equals the census wage bill per paid worker across manufacturing, trade and services, which amounts to 165.4 thousand 2023 pesos in the 2019 census and 168.7 thousand in 2024. The labor force $\bar{L}$ is the economically active population of the ENOE, averaged over the four quarters of 2018 and of 2023, 55.6 and 60.6 million people, so that the non-manufacturing sector stands for the rest of the economy. At that wage, manufacturing employs whatever labor its firms demand (17), and the competitive non-manufacturing sector employs everyone else.[9] Lastly,

[9]It is important to mention that the counterfactual aggregates of Section 6 are evaluated at the firm choices of the executed solve, with the domestic demand loop (35) solved again under this labor force. This means that firms are not re-optimized against the new demand, an omission that shows to be second order, as even at the largest surcharge it stays below 0.02 percentage points of real income. I also assume that intermediate purchases scale with manufacturing sales.

Figure 3: Trade-weighted CDFs of the policy parameters μ_p^{CTC} and μ_p^{RVC} under NAFTA and USMCA



Notes: 2018 weights for NAFTA, 2023 for USMCA.

$\chi = 0.37$ is the share of Mexican final consumption that is spent on manufactures, which I take from the INEGI use table. It only weights the manufacturing price index in the welfare deflator (19) and plays no role in the demand loop (35).

Compliance rate. The compliance rate over all claims is the share of U.S. & CA-bound revenue that is shipped under preference, $\sum_{f,p} d_{fp} R_{f,p,\text{U.S. \& CA}} / \sum_{f,p} R_{f,p,\text{U.S. \& CA}}$, where d_{fp} is the optimal choice in (14). Measured in this way, the model rate can be compared directly to the empirical benchmark in Table 3. Products that do not have a coded rule (1.2% of U.S. & CA-bound revenue in the 2018 panel and 1.7% in the 2023 panel) are never able to comply, and they do not pay a tariff either. Additionally, a zero-NTR product whose rule does not bind is able to comply at no cost at all, so the compliance rate that I report is the *strict* one, under which a product only counts as compliant if it faces a positive NTR rate. The data counterpart of this measure, the value claimed on positive-NTR lines over all value, is 52.3% in 2018 and 46.5% in 2024.

4.3 Structural Estimation

The remaining parameters are the firm-level sourcing-efficiency and market-reach vectors $\mathbf{a}_f = (a_{f,s})_{s \neq \text{U.S. \& CA}}$ and $\mathbf{b}_f = (b_{f,d})_{d \in \mathcal{D}}$, and the location parameters of the per-source and per-destination fixed-cost distributions $\{\mu_s, \mu_d\}$. The strategy consists in recovering $(\mathbf{a}_f, \mathbf{b}_f)$ for every firm by inverting the model on the cells where the firm is observed, and predicting them from a Heckman-corrected regression on the cells where it is not. I then estimate the fixed-cost location vector by inverting the model on aggregate participation shares, with one moment per parameter. The normalizing coordinate is $a_{f,\text{U.S. \& CA}} \equiv 1$, given that U.S. & CA is the source block with the highest participation (Table 7), meaning that the ratio denominator $\mu_{f,\text{U.S. \& CA}}$ is non-zero for the largest possible subsample of firms. No normalization is imposed on the demand side, as $b_{f,d}$ is inverted from (22) at all three destinations and its imputation regression is run on $\log b_{f,d}$ directly.

Firm primitives ($\mathbf{a}_f$, $\mathbf{b}_f$): inversion on observed cells. On the cells where firm f actually sources from s or sells to d, the primitives can be backed out directly from the model. Log-inverting the Fréchet share equation (7) at $a_{f,\text{U.S. \& CA}} \equiv 1$ gives the sourcing efficiency, while summing the CES revenue equation (13) across the firm's products at destination d and solving for the market reach gives:

$$\log a_{f,s} = \tfrac{1}{\theta}\left[\log(\mu_{f,s}/\mu_{f,\text{U.S. \& CA}}) - \log(T_s(c_s\tau_s)^{-\theta})\right], \tag{21}$$

$$b_{f,d}^{\sigma-1} = \frac{\sum_p R_{f,p,d}}{E_d\, P_d^{\sigma-1} \sum_{p \in \mathcal{P}_f} \varpi_{f,p}\, p_{f,p,d}(\mathcal{S}_f, d_{fp})^{\,1-\sigma}}, \tag{22}$$

where (E_d, P_d) is the market bundle of destination d, Mexico included, set as in Section 4. The inversions are exact on $\{f : \mu_{f,s} > 0,\ \mu_{f,\text{U.S. \& CA}} > 0\}$ and on $\{f : Y_{f,d} > 0\}$, and they are not defined anywhere else. This leaves a lot of empty cells, and how I fill them matters. With zeros, no firm could ever start sourcing from a new country or selling in a new destination after a policy shock, so the extensive margin would be gone from the counterfactuals. What I use instead is the prediction of a Heckman-corrected regression.

Firm primitives ($\mathbf{a}_f$, $\mathbf{b}_f$): Heckman two-step on missing cells. Running OLS of the mean functions in (21) and (22) on the selected subsample would give estimates that are biased upward. Firms do not source from s at random. The ones that do are the ones with a high latent $a_{f,s}$, so the fitted coefficients would overstate the unconditional expectation. My approach is to put the firm–source pairs of the five source blocks other than U.S. & CA in one long panel and fit a single Heckman (1979) two-step regression on it for the latent sourcing efficiency (U.S. & CA is left out because it is the normalizing coordinate, so there is nothing to impute there). For the latent market reach I do the same, this time stacking the three

destinations. Both regressions also pool 2018 and 2023, with a year fixed effect in both stages. The two equilibria therefore share one imputation surface up to a level shift, and the imputed cells of a given period are not estimated from that period's sample alone. On the sourcing side the panel is the five blocks times the firms with $\mu_{f,\text{U.S. \& CA}} > 0$, since the normalized log-ratio is only defined for them (4,947 firms in 2018 and 4,462 in 2023). That is 47,045 firm–block rows, and the remaining 779 and 686 firms get fitted values. On the demand side it is the three destinations times the exporters of both periods, meaning the 9,389 firms with U.S. & CA or RoW sales, for 28,167 rows. Import-only firms are not part of the demand panel. They sell nothing abroad by construction, and the selection equation would mistake them for firms that decided to stay out of both foreign markets, so I give them fitted values on their foreign coordinates instead. Helpman et al. (2008) use a similar correction for the selection of firms into export markets, and here I also apply it to the selection into source markets.

$$\text{Stage 1 (selection):} \quad \Pr(\mu_{f,s} > 0 \mid X_{f,s}, W_f) = \Phi(X'_{f,s}\delta_a + \delta_a^W W_f), \tag{23}$$

$$\text{Stage 2 (outcome):} \quad \log(\mu_{f,s}/\mu_{f,\text{U.S. \& CA}}) = X'_{f,s}\beta_a + \lambda_a \,\text{IMR}(X'_{f,s}\hat{\delta}_a + \hat{\delta}_a^W W_f) + u_{f,s}\,, \tag{24}$$

where the regressor vector of the outcome stage $X_{f,s}$ includes firm size ($\log L_f$), source-block fixed effects, ownership-country fixed effects, NAICS-3 sector fixed effects and the year fixed effect. W_f is the *exclusion restriction* and only enters the selection stage. On the sourcing side the exclusion is the number of establishments of the firm ($\log N_{estab}$), on the demand side the breadth of its product range ($\log N_{HS6}$). Table 8 reports the first-stage coefficient on each of them. $\text{IMR}(z) = \phi(z)/\Phi(z)$ is the inverse Mills ratio. Differences across sources are captured by the source fixed effects, since all of the blocks go into one regression and these fixed effects absorb the mean gap in log shares between them.

The choice of exclusions follows the distinction that the model makes between the extensive and the intensive margin. On the sourcing side I use the number of establishments. The idea behind this is that a firm with several plants can spread the fixed cost of a relationship with a foreign supplier (procurement staff, customs registration, supplier search) over a larger operation, which may make this cost easier to pay. This makes it more likely to source from a given country, but once it sources from s the share equation (7) is scale-free and depends only on $a_{f,s}$. On the demand side I use the breadth of the product range, under the notion that a firm with more products obtains a larger variable profit from paying the entry cost of a destination once, whereas for a firm that already serves d the inversion (22)

nets the product mix out of $b_{f,d}$. Breadth should therefore not move market reach itself. The identifying assumption is that, conditional on the controls, the number of establishments and the breadth of the product range only move the capacity of the firm to pay fixed costs and are not proxies for its latent sourcing efficiency or market reach. If $(\varepsilon_{f,s}, u_{f,s})$ is bivariate normal, the linear-IMR form is the closed-form correction, with $\lambda_a = \rho_a \sigma_u$, and the fitted $\hat{\beta}_a$ then recovers the bias-corrected unconditional mean $X'_{f,s}\hat{\beta}_a = \mathbb{E}[\log(\mu_{f,s}/\mu_{f,\text{U.S. \& CA}}) \mid X_{f,s}]$. Plugging this into (21) gives the imputed $\hat{a}_{f,s}$ on the missing cells:

$$\log \hat{a}_{f,s} = \tfrac{1}{\theta}\left[X'_{f,s}\hat{\beta}_a - \log(T_s(c_s\tau_s)^{-\theta})\right] \quad \text{for } \mu_{f,s} = 0 \text{ or } \mu_{f,\text{U.S. \& CA}} = 0,$$

whereas the closed-form inversion (21) is kept for the observed cells. Following the same procedure on the pooled firm–destination panel gives $\hat{b}_{f,d}$. On both sides I cap the imputed cells at the largest value inverted from data in the same block or destination. The result is a firm-by-source matrix $\hat{\mathbf{A}}$ and a firm-by-destination matrix $\hat{\mathbf{B}}$. These are what let firms enter new sources and destinations in the counterfactuals, since an inactive cell can become active after a policy shock whenever its imputed value is high enough to clear the fixed cost.

Fixed costs (κ_s, κ_d): moment inversion on aggregate participation shares with lognormal draws. Firm-country and firm-destination fixed costs are drawn iid from log-normal distributions, following Antràs et al. (2017) and Morán and Cebreros (2025):

$$\kappa_{f,s} \sim \text{LogNormal}(\mu_s,\, \delta), \qquad \kappa_{f,d} \sim \text{LogNormal}(\mu_d,\, \delta), \tag{25}$$

with one draw per firm per source country and one per firm per destination. κ is the fixed cost that enters the firm's profit function (14), and μ is the log-space location parameter of its distribution, with $E[\kappa_c] = \exp(\mu_c + \delta^2/2)$. I pin the dispersion at $\delta = \sqrt{\log 3}$, following Morán and Cebreros (2025).

Following Antràs et al. (2017), I pin $\kappa_s^{MX} \equiv 0$, given that every Mexican manufacturer sources domestically by construction. Even if a firm sources no material inputs domestically, every firm sources at least some labor domestically, so there is no meaningful *entry* fixed cost into the Mexican input market. The μ of the remaining eight blocks (five non-MX sources and three destinations) are estimated. The estimation recovers the eight location parameters

$$\boldsymbol{\mu} = \left\{\mu_s^{\text{U.S. \& CA}},\, \mu_s^{AX},\, \mu_s^{CH},\, \mu_s^{EU},\, \mu_s^{RoW},\, \mu_d^{\text{U.S. \& CA}},\, \mu_d^{MX},\, \mu_d^{RoW}\right\},$$

one per estimated block. The results table reports $\hat{\kappa}_c = \exp(\hat{\mu}_c + \delta^2/2)$, and the rest of the paper refers to fixed costs as κ.

Model solution for a given $\boldsymbol{\mu}$. For a given $\boldsymbol{\mu}$, every firm has its own draw $\kappa_{f,c} = \exp(\mu_c +$

$\delta\, z_{f,c}$) in each of the nine blocks ($\kappa_{f,s}^{MX} = 0$ is pinned), where the standard normals $z_{f,c}$ are drawn once and kept fixed across evaluations so that the simulated moments use common random numbers. Solving (14) once per firm at these draws then gives each firm's sourcing set $\mathcal{S}_f$ and destination set $\mathcal{D}_f$.

Moment vector. Each of the eight location parameters is matched to one participation moment, with the five non-MX source-side fixed costs identified from the extensive margin of sourcing,

$$m_s^{\text{source}} = \Pr\big(\mu_{f,s} > 0\big), \quad s \in \{\text{U.S. \& CA}, AX, CH, EU, RoW\}.$$

The other three identify the destination-side fixed costs through the extensive margin of selling into each destination:

$$m_d^{\text{dest}} = \Pr\big(Y_{f,d} > 0\big), \quad d \in \{\text{U.S. \& CA}, MX, RoW\}.$$

Additional moments are computed but not targeted, and I use them as checks in Section 5.1. They include the aggregate compliance rate (the share of MX→US import value claimed under preference, from USA Trade Online) and the share of firms sourcing from at least two foreign blocks, which depends on how participation is correlated across blocks and is therefore not pinned down by the marginal shares.

Estimator. With $\hat{\boldsymbol{m}}(\boldsymbol{\mu})$ the eight simulated participation rates at $\boldsymbol{\mu}$, $\boldsymbol{m}$ their data counterparts and $\hat{\boldsymbol{g}}(\boldsymbol{\mu}) = \hat{\boldsymbol{m}}(\boldsymbol{\mu}) - \boldsymbol{m} \in \mathbb{R}^8$ the gaps between them, the estimator is the root

$$\hat{\boldsymbol{\mu}}: \quad \hat{\boldsymbol{g}}(\hat{\boldsymbol{\mu}}) = \mathbf{0}. \tag{26}$$

Participation in block c is weakly decreasing in μ_c, given that a higher μ_c raises $\kappa_{f,c}$ for every firm, which can only take firms out of the block, meaning that each gap is monotone in its own parameter. Having eight parameters and eight monotone moments, (26) is exactly identified, which allows one to solve it by inversion instead of minimizing a distance. This corresponds to the just-identified case of the simulated method of moments, where the moment conditions (26) hold with equality up to the $1/N$ step of the simulated shares. Recovering parameters through an exact inversion, with one moment per parameter, is common practice in quantitative trade models. To state an example, Melitz and Redding (2015) choose the fixed exporting cost so as to reproduce the observed share of exporting firms, and, as Redding and Rossi-Hansberg (2017) note for the class of exactly identified models that grew out of Eaton and Kortum (2002), observed outcomes map one-to-one into the unobserved primitives that rationalize them as an equilibrium.

5 Estimation Results

Section 5.1 reports the structural estimates and the fit of the model to targeted and untargeted moments in both periods. Section 5.2 then examines the estimated firm primitives in 2023, while Appendix E reports the same objects for 2018.

5.1 Parameter Estimates and Model Fit

Table 6: Structural parameter estimates, both periods

	2018		2023	
Parameter	$\hat{\kappa}$	SE_κ	$\hat{\kappa}$	SE_κ
Sourcing fixed costs κ_s				
U.S. & CA	3.220	0.366	3.385	0.342
Asia excl. China	0.057	0.026	0.080	0.032
China	0.045	0.020	0.057	0.026
Europe	0.053	0.023	0.075	0.032
Rest of World	0.042	0.017	0.062	0.025
Destination entry fixed costs κ_d				
U.S. & CA	7.89	2.81	9.08	2.34
Mexico	30.59	6.77	28.94	4.64
Rest of World	4.56	1.83	4.45	1.14
Firms	5,726		5,148	
firm×HS6 products	136,762		122,631	

Notes: $\hat{\kappa}$ is the mean of the fitted distribution, $\exp(\hat{\mu}+\delta^2/2)$. SE_κ is the standard deviation of $\hat{\kappa}$ (not of $\hat{\mu}$) across $B = 100$ firm-cluster bootstrap replicates. Fixed costs are measured in Mexican labor, as worker-years at the pinned wage, and $\kappa_s^{\text{MX}} \equiv 0$.

Table 6 reports the estimates, and two things about them are worth pointing out. The first has to do with the size of the fixed costs. Sourcing from U.S. & CA shows to be by far the most expensive option in both periods, at 3.2 to 3.4 worker-years, while no other source block costs more than a tenth of a worker-year. Between the two periods the U.S. & CA cost rises by 5% and the others by more, although from very low levels: +39% for Asia excluding China, +26% for China, +43% for Europe and +48% for the Rest of World. On the destination side the domestic market is by far the most expensive, at about 30 worker-years in both periods. Almost every firm has a high inverted domestic market reach, and a large entry cost is the only way to square that with the 11 to 14% of firms that do not sell at home. The U.S. & CA entry cost rises by 15%, while the domestic and Rest-of-World costs fall by 5 and 2%. In terms of revenue these fixed costs are modest, with domestic entry, the largest, at about 0.2% of mean firm revenue and U.S. & CA sourcing at about 0.02%. Second, the

eight location parameters are all interior, with bootstrap intervals well inside the inversion bracket, and the inversion hits every participation target (Table 7).

Model fit. The inversion matches all eight targeted participation shares (Table 7). Among the untargeted moments, the share of firms sourcing from at least two foreign blocks is close to the data (87.9% against 86.6% in 2018, 89.6% against 87.4% in 2023), the U.S. & CA share of revenue (44% and 42%) matches, and the value-weighted composition of materials by source is within 0.3 percentage points of the data on every block. Where the model does worse is on the compliance rate, which it overpredicts in both periods (58.2% against 52.3% in 2018 and 57.1% against 46.5% in 2023). It is important to mention that on the model side zero-NTR products count as non-compliant, even though they could comply at no cost, and that over all claims the rates in the data are 56% and 48%. My reading of this gap is that claiming preference carries a cost that is not in the model. In line with this, once the 2025 surcharge made not complying expensive, compliance in the data went from 44% to 85% (Table 3). For this reason I trust the counterfactuals more on how firms respond to policy than on the level of utilization at low tariffs.

Selection-corrected imputation. The outcome stage of the two pooled Heckman regressions is in Table 8. On both sides the inverse-Mills loading comes out positive, at 3.51 (s.e. 0.36) for sourcing efficiency and a smaller 0.44 (s.e. 0.18) for market reach. So selection goes in the direction one would expect. A firm that sources from a block, or sells in a market, has a higher latent sourcing efficiency or market reach than its observables alone would predict. Without the correction, OLS on the observed cells would overstate the latent means, and indeed the unconditional fit that I use for the missing cells lies below the observed cells. Note that the sourcing-efficiency surface is estimated on the log share ratio, i.e. on $\theta \log a_{f,s}$ up to a block constant (21). It increases with firm size and is 0.20 higher in 2023, or 0.11 once converted to $\log a_{f,s}$.

Estimated firm primitives. In Table 9 I report the distribution across firms of $\log a_{f,s}$ and $\log b_{f,d}$ that results from the inversions and from the Heckman-corrected fits that I described above, together with the share of firms for which each coordinate is inverted from the data rather than imputed. The sourcing efficiencies are widely spread, with standard deviations of 2.0 to 2.1 log units, and there is a clear ordering across sources. Mexico has the highest mean, the CH, EU and AX blocks cluster around a common lower mean and RoW is below all of them. The demand coordinates follow the same order as in the data, Mexico first, with the Rest of World, a market that just over half of the firms serve, about 1.6 to 2.4 log units below the other two.

Table 7: Targeted and untargeted moments: data vs. model, both periods

	2018			2023		
Moment	Data	Model	Diff	Data	Model	Diff
Source-side participation shares $\Pr(\mu_{f,s} > 0)$ *(targeted)*						
$s =$ U.S. & CA	0.864	0.864	−0.000	0.867	0.867	−0.000
$s =$ AX	0.739	0.739	−0.000	0.761	0.761	−0.000
$s =$ CH	0.784	0.784	−0.000	0.814	0.814	−0.000
$s =$ EU	0.759	0.760	−0.000	0.770	0.770	+0.000
$s =$ RoW	0.646	0.645	+0.000	0.650	0.651	−0.000
Destination-side participation shares $\Pr(Y_{f,d} > 0)$ *(targeted)*						
$d =$ U.S. & CA	0.780	0.780	+0.000	0.798	0.798	+0.000
$d =$ MX	0.860	0.860	−0.000	0.892	0.892	+0.000
$d =$ RoW	0.528	0.528	+0.000	0.556	0.556	+0.000
Compliance (untargeted)						
Compliance rate	0.523	0.582	−0.059	0.465	0.571	−0.106
Revenue share by destination (untargeted)						
U.S. & CA	0.444	0.445	−0.001	0.417	0.423	−0.006
Mexico	0.494	0.494	+0.000	0.527	0.522	+0.005
RoW	0.061	0.062	−0.000	0.055	0.055	−0.000
Share of materials by source (untargeted)						
U.S. & CA	0.328	0.328	−0.000	0.336	0.336	−0.000
Mexico	0.367	0.369	−0.002	0.325	0.328	−0.003
Asia excl. China	0.116	0.115	+0.001	0.122	0.120	+0.002
China	0.096	0.095	+0.001	0.102	0.101	+0.001
Europe	0.061	0.061	+0.000	0.070	0.070	+0.000
Rest of World	0.032	0.032	+0.000	0.045	0.044	+0.000

Notes: Diff = Data − Model. Compliance rate: under NAFTA rules in 2018 and USMCA rules in 2023 (Section 4.2). Revenue and material shares: means of firm shares weighted by census revenue and census materials, at each period's baseline solve (2018 under NAFTA rules).

5.2 Validating the Estimated Firm Primitives

This subsection reports the 2023 period throughout, while Appendix E reports each object for the 2018 period.

Ownership and the geography of the primitives. Tables 10a and 10b split the two blocks of Table 9 by the ownership country of the firm for 2023. Perhaps the most interesting observation on the sourcing side is that the geography lines up with ownership. Whenever the home region of a group is part of the sourcing menu, that is the group's best coordinate (the bold diagonal of Table 10a). Mexican-owned firms have their highest sourcing efficiency in Mexico (+1.00), AX-owned firms in Asia excluding China (+1.54, which is the largest entry in the panel), Chinese-owned firms in China (+1.33) and EU-owned firms in Europe (+0.71).

Table 8: Heckman second stage: sourcing-efficiency and market-reach imputation equations

	Sourcing efficiency $\log(\mu_{f,s}/\mu_{f,\text{U.S. \& CA}})$	Market reach $\log b_{f,d}$
$\log L_f$	0.315	0.239
	(0.033)	(0.011)
Inverse Mills ratio ($\hat{\lambda}$)	3.505	0.442
	(0.356)	(0.185)
Constant	−2.271	3.568
	(0.450)	(0.092)
First stage (selection probit)		
Exclusion: $\log N_{estab}$	0.048	
	(0.011)	
Exclusion: $\log N_{HS6}$		0.085
		(0.008)
Source FE	Yes	
Destination FE		Yes
Ownership FE	Yes	Yes
Sector (NAICS-3) FE	Yes	Yes
Year FE	Yes	Yes
N (panel)	47,045	28,167
N (selected)	36,424	22,510

Notes: Heckman (1979) two-step estimated on the 2018 and 2023 panels stacked together and pooled across blocks, with a year fixed effect in both stages. Standard errors in parentheses. Ownership fixed effects are relative to Mexico, six groups on the sourcing side (the source blocks) and three on the demand side (Mexico, U.S. & CA and other foreign). Each exclusion enters the selection probit of its own side only. The sourcing-efficiency panel consists of the firms with $\mu_{f,\text{U.S. \& CA}} > 0$ times five blocks, and the demand panel of the exporters (firms with U.S. & CA or RoW sales) times three destinations.

For U.S. & CA-owned firms the home block is the normalizing one ($a_{f,\text{U.S. \& CA}} \equiv 1$), and they lie below zero on every other block, so for them too the home block is the best. Notably, the peaks of the AX- and CH-owned firms are 2.5 to 3.2 log points above the means of the Mexican-owned firms in those same blocks. The RoW-owned group, which pools several origins, is the only one that does not peak in its own block. On the demand side the pattern holds for the two North American markets (Table 10b), with Mexican-owned firms having the highest mean market reach in Mexico (6.1) and U.S. & CA-owned firms in the U.S. & CA market (6.2), whereas in the Rest of World the remaining foreign owners (3.9) are level with the U.S. & CA-owned firms (3.9).

Note how these differences in levels translate directly into what firms do. Through the lens of the model, a higher $a_{f,s}$ makes a firm more likely to pay the fixed cost of block s and, once it does, to spend a larger share $\mu_{f,s}$ on that block, while a higher $b_{f,d}$ raises the revenue of the firm in market d. Foreign affiliates thus source more from the home regions of their parents and sell more there as well. Note that this pattern comes out of the inverted cells.

Table 9: Estimated firm primitives: cross-firm distribution (2023)

Coordinate	mean	sd	observed
Sourcing efficiency $\log a_{f,s}$ *(U.S. & CA* $\equiv 0$*)*			
Mexico	−0.115	2.137	0.55
Asia excl. China	−1.133	2.005	0.72
China	−0.945	2.014	0.75
Europe	−0.978	2.004	0.72
Rest of World	−1.558	2.084	0.62
Market reach $\log b_{f,d}$			
U.S. & CA	5.216	1.747	0.80
Mexico	5.987	1.469	0.89
Rest of World	3.573	1.598	0.56

Notes: 5,148 traders. Direct inversion on observed cells; Heckman-fitted mean on missing cells, capped at the largest inverted value of the coordinate. "observed": share of firms with the coordinate inverted from data; a sourcing-efficiency coordinate counts as observed only for firms that also source from U.S. & CA, the normalizing block. $\log a_{f,\text{U.S. \& CA}} \equiv 0$. Means and standard deviations aggregate the ownership-group statistics of the replication workbook by firm counts. 2018 in Appendix E.

Table 10a: Estimated sourcing efficiency by ownership group (2023)

	Ownership group					
	Mexico	U.S. & CA	Asia excl. China	China	Europe	Rest of World
Firms	2,647	1,475	227	23	515	261
Mean sourcing efficiency $\log a_{f,s}$*, by source block (U.S. & CA* $\equiv 0$*)*						
Mexico	**1.00**	−1.95	0.26	−0.39	−0.52	−0.56
U.S. & CA	0.00	**0.00**	0.00	0.00	0.00	0.00
Asia excl. China	−1.65	−0.87	**1.54**	0.51	−0.66	−0.81
China	−1.19	−0.86	−0.08	**1.33**	−0.46	−0.83
Europe	−1.13	−1.39	−1.23	−1.02	**0.71**	−0.27
Rest of World	−1.64	−1.87	−2.06	−1.46	−0.39	**−0.81**

Notes: Means over all firms in each group, observed and Heckman-imputed cells together; per-coordinate observed shares in Table 9. Bold: each group's home block. $\log a_{f,\text{U.S. \& CA}} \equiv 0$ for every firm, so the U.S. & CA row is zero and each entry is a sourcing efficiency relative to U.S. & CA sourcing. Owners from Asia excl. China are those from Japan, Korea and Singapore, the Asian origins coded in both censuses; owners from other Asian countries are in the Rest of World group. 2018 in Appendix E.

Variation explained by observables. Table 11 reports the share of the cross-firm variation in each $\log a_{f,s}$ and $\log b_{f,d}$ latent that is explained by ownership-country and industry (NAICS-3) fixed effects and their interaction. Let the primitive coordinates be indexed by k, which is a source block s on the sourcing side or a destination d on the demand side, and let $x_{f,k}$ be firm f's estimated value of coordinate k, that is, $x_{f,k} = \log a_{f,s}$ for $k = s$ and $x_{f,k} = \log b_{f,d}$ for $k = d$, inverted or imputed as in Table 9. Let $o(f)$ be the firm's ownership

Table 10b: Estimated market reach by ownership group (2023)

	Ownership group		
	Mexico	U.S. & CA	Other foreign
Firms	2,647	1,475	1,026
Mean market reach $\log b_{f,d}$*, by destination*			
Mexico	**6.09**	5.91	5.84
U.S. & CA	4.65	**6.24**	5.21
Rest of World	3.28	3.90	**3.86**

Notes: Means over all firms in each group, observed and Heckman-imputed cells together. Owners are grouped by destination block, so "Other foreign" pools the firms owned from Asia excl. China, China, Europe and the Rest of World in Table 10a. Bold: each group's home market. 2018 in Appendix E.

group and $n(f)$ its NAICS-3 sector. For each coordinate, I estimate four regressions by OLS,

$$\begin{aligned} x_{f,k} &= \xi^{o}_{k,o(f)} + e^{\text{own}}_{f,k}, & x_{f,k} &= \xi^{n}_{k,n(f)} + e^{\text{ind}}_{f,k}, \\ x_{f,k} &= \xi^{o}_{k,o(f)} + \xi^{n}_{k,n(f)} + e^{\text{both}}_{f,k}, & x_{f,k} &= \xi^{on}_{k,o(f),n(f)} + e^{\times}_{f,k}, \end{aligned} \tag{27}$$

where $\xi^{o}_{k,o}$ is a fixed effect for ownership group o, $\xi^{n}_{k,n}$ is a fixed effect for sector n, $\xi^{on}_{k,o,n}$ is a fixed effect for each ownership–sector cell, all of them specific to coordinate k, and $e^{m}_{f,k}$ is the residual of specification m. The table reports the R^2 of each of these regressions, and of two more that add firm size, $\log L_f$, alone and on top of the full interaction, since size is the continuous regressor of the Heckman mean function. The goal of this exercise is to address a natural question regarding the Heckman imputation, which is how much of the heterogeneity in $(\mathbf{a}_f, \mathbf{b}_f)$ is picked up by the coarse observables (ownership and sector) that go into the mean function, and how much of it is idiosyncratic. Even when they are fully interacted, the observables explain at most 43% of the variation across firms in any of the primitive coordinates, and adding firm size takes this to at most 49%. Ownership shows to be the stronger of the two for the MX, EU and RoW sourcing-efficiency coordinates and ties with sector for AX, whereas sector is the stronger one for China and for all three demand coordinates. It is important to mention that the R^2 are biased upward, given that the imputed cells are fitted values of a mean function that contains these same ownership and sector dummies. Even so, the conclusion that observables account for a minority of the variation holds in spite of this bias.

Cross-country correlation structure of the primitives. Tables 12a and 12b report the sample correlation of $(\log a_f, \log b_f)$ across the 2023 firms. An interesting observation in the sourcing-efficiency block (Table 12a) is that there is a clear East-Asian cluster, given that the AX–CH correlation of 0.51 is the largest pairwise correlation, above the -0.09 to 0.46

Table 11: Share of the cross-firm variation in $\log a_{f,s}$ and $\log b_{f,d}$ explained by ownership and sector (2023)

		R^2 from fixed effects for				Adding firm size $\log L_f$	
Latent	Coordinate	Own	Sector	Own + Sector	Own × Sector	Alone	+ Own × Sector
$\log a$	MX	0.358	0.174	0.411	0.434	0.019	0.435
	AX	0.127	0.132	0.206	0.219	0.050	0.248
	CH	0.028	0.098	0.111	0.128	0.028	0.152
	EU	0.093	0.053	0.135	0.153	0.074	0.215
	RoW	0.048	0.033	0.080	0.099	0.077	0.171
$\log b$	U.S. & CA	0.152	0.339	0.411	0.423	0.094	0.491
	MX	0.005	0.207	0.214	0.223	0.036	0.280
	RoW	0.037	0.249	0.276	0.287	0.059	0.340

Notes: 5,148 traders. R^2 from OLS on ownership dummies alone, NAICS-3 dummies alone, their union, and full interaction, equation (27), and from two more regressions that add firm size ($\log L_f$, the size regressor of the Heckman mean function) alone and on top of the full interaction. Ownership has six levels for the sourcing-efficiency rows and three levels (Mexico, U.S. & CA, other foreign) for the market-reach rows. 2018 in Appendix E.

range observed for the other pairs of blocks. Firms with an above-average sourcing efficiency in China tend to have an above-average sourcing efficiency in AX as well, which is consistent with a latent "regional supplier-network access" factor. This correlation helps explain part of the sourcing-reallocation pattern in Section 6.5, where sourcing shifts toward Asia excluding China about twice as much, in proportion, as toward any other block when China becomes more expensive. The demand block (Table 12b) shows positive and moderate correlations, at 0.52 for U.S. & CA–RoW, 0.43 for Mexico–RoW and 0.35 for U.S. & CA–Mexico, meaning that firms with a high market reach in one foreign market tend to have it in the other one as well, rather than specializing in one of them.

Table 12a: Correlation structure of estimated sourcing efficiency $a_{f,s}$ (2023)

	MX	AX	CH	EU	RoW
MX	1.00				
AX	−.09	1.00			
CH	.01	**.51**	1.00		
EU	.15	.35	.34	1.00	
RoW	.09	.28	.27	.46	1.00

Notes: 5,148 traders. Lower triangle of the correlation matrix (upper triangle omitted by symmetry). U.S. & CA is omitted ($\log a_{f,\text{U.S. \& CA}} \equiv 0$). 2018 in Appendix E.

Table 12b: Correlation structure of estimated market reach $b_{f,d}$ (2023)

	U.S. & CA	MX	RoW
U.S. & CA	1.00		
MX	.35	1.00	
RoW	.52	.43	1.00

Notes: 5,148 traders. Lower triangle of the correlation matrix (upper triangle omitted by symmetry). 2018 in Appendix E.

6 Counterfactual Analysis

The two equilibria that I calibrate bracket the trade war and the switch from NAFTA to USMCA, while the 2025 surcharge came after both of them and raised the cost of staying outside of the agreement. The structure of this section follows that same chronology. I begin by performing a Shapley decomposition of the 2018–2023 transition in order to understand what moved between the two equilibria (Section 6.1). I follow by looking at the surcharge (Section 6.2) and at what the agreement was worth once the surcharge was in place (Section 6.3). Lastly, I turn to the policies that are on the table right now, tighter content rules at the observed and at the 2025 tariffs (Section 6.4) and tariffs on Chinese inputs (Section 6.5). It is important to mention that every policy experiment is a comparative static of the static model. In each of them firms are allowed to re-optimize their sourcing and destination sets, their compliance choices and their quantities, and the Mexican demand loop is solved again, whereas the firm primitives, the fixed-cost draws, the wage and the set of firms (apart from exit) stay at their baseline values. The experiment then changes the cost of the option that was the cheapest for a firm at baseline, given that a tighter rule of origin raises the cost of complying, whereas a higher tariff raises the cost of not complying. Each firm then re-optimizes, meaning that responses are heterogeneous: for some firms not complying remains the cheapest option, for others complying becomes the cheapest option, and some others keep complying at a higher cost. The aggregate patterns reported below summarize these firm-level choices. In every counterfactual, the foreign pairs (E_d, P_d) are held fixed and the Mexican pair (E_M, P_M) moves with income through the demand loop (35) (Section 3.4).

6.1 Two-Equilibrium Shapley Decomposition

I calibrate the model at two different points in time. Equilibrium E_A is based on CE 2019 and PEME 2018 with NAFTA rules of origin and pre-trade-war conditions, whereas equilibrium E_B is based on CE 2024 and PEME 2023 with USMCA rules and post-trade-war conditions. With these two equilibria at hand I perform a Shapley decomposition to understand the

effects that different policy, firm-level and macroeconomic channels had over the transition in this period on different model outcomes. The usefulness of this methodology comes from being able to attribute the non-linear outcomes that come from the interaction of various factors in structural models such as the one presented in this paper. Audoly et al. (2025) have recently highlighted its usefulness for exactly this kind of exercise, and it has been used in quantitative trade (Allen and Arkolakis, 2014) and in macroeconomic analysis (Paz-Pardo, 2024).

Let $\mathcal{K}$ be the set of the $m = 10$ inputs that differ between E_A and E_B, which Table 21 in Appendix F lists in three groups: the policy, the firm panel and the aggregates. For any subset $\mathcal{J} \subseteq \mathcal{K}$, $X(\mathcal{J})$ denotes the outcome of the model solved with the factors in $\mathcal{J}$ at their 2023 values and the rest at their 2018 values, so that $X(\emptyset) = X_A$ and $X(\mathcal{K}) = X_B$. The contribution of factor k is its Shapley value (Shapley, 1953, Shorrocks, 2013),

$$C_k \;=\; \sum_{\mathcal{J}\subseteq\mathcal{K}\setminus\{k\}} \frac{|\mathcal{J}|!\,(m-|\mathcal{J}|-1)!}{m!}\big[X(\mathcal{J}\cup\{k\}) - X(\mathcal{J})\big], \qquad \sum_{k\in\mathcal{K}} C_k \;=\; X_B - X_A, \quad (28)$$

that is, the change in X from switching factor k to its 2023 value, averaged over every combination $\mathcal{J}$ of the other factors that could already be at their 2023 values. By construction the contributions add up exactly to $X_B - X_A$, meaning that there is no interaction term or residual left over, and no need to choose an order in which to switch the factors. The details of the implementation are in Appendix F.

Table 13 reports the decomposition. The first result to highlight is that the implementation of USMCA, which tends to have stricter rules of origin had a small impact on overall welfare, showing it accounts for a drop of -0.07% of real income. This speaks to the relatively low average tariffs observed in 2023, meaning that more firms simply opt to pay the tariff instead of taking the free trade agreement route. In the case of compliance, however, the rule switch shows to be the largest single factor, lowering compliance by 2.41 points against an aggregate fall of 1.18. Firm turnover ends up masking a good part of this, given that the firms that exited were less compliant than the ones that remained (+3.47), entrants add 1.35 and the re-estimated primitives of the continuing firms subtract 2.75. On the sourcing side, perhaps the clearest sign of a larger Chinese presence following the U.S.–China trade war is that the change in sourcing conditions alone raised intermediate purchases from China by 14.79%, an increase that is larger than that of any other source country and larger than the one coming from any other macroeconomic variable. The same shift in sourcing conditions also raised purchases from the Rest of World (13.50) and from Europe (11.54), whereas it lowered those from U.S. & CA (-1.77). Through the lens of the model, then, the trade-war period looks like a drift toward inputs from outside of North America. Foreign demand, on

the other hand, raised purchases from every single source by 7 to 9 points, and it is the main factor behind the rise in purchases from U.S. & CA, whereas the larger labor force is the one behind domestic sales. Sales to U.S. & CA and to the Rest of World fell by 4.97 and 7.91%. This comes from the firm side (−19.20 and −24.77, with the re-estimated primitives of the continuing firms alone at −20.03 and −26.75), which more than offsets foreign demand (+11.11 and +12.22).[10] The rules account for only −0.32 and 0.00. The rule switch mattered so little for income because the tariff it exempts firms from was low in both years, and raising that tariff is exactly what the next exercise does.

Table 13: Shapley decomposition of the 2018–2023 transition

		Policy	Firm			Aggregates						
Outcome	Total	R	$\hat{a}, \hat{b}$	E	X	ψ	κ	D	A_o	$\bar{L}$	Θ	Gap
Aggregate outcomes												
Compliance rate (pp)	−1.18	−2.41	−2.75	1.35	3.47	−1.16	0.00	0.00	0.10	0.00	0.00	0.21
Real income W_M	11.34	−0.07	−5.01	4.11	−3.86	1.04	0.00	2.69	1.42	7.15	3.84	0.02
Profits, traders	0.43	−0.18	−14.28	13.73	−12.02	3.29	0.00	7.10	0.19	3.40	−0.70	−0.11
Intermediate purchases by source $M_{MX,s}$												
U.S. & CA	21.37	−0.11	7.23	13.04	−8.24	−1.77	0.00	8.19	0.30	3.39	−0.68	0.03
Mexico	−11.49	−0.17	−30.18	17.72	−10.55	2.72	0.00	5.12	0.67	4.38	−0.91	−0.29
Asia excl. China	−3.91	−0.31	−11.93	6.60	−12.37	4.22	0.00	8.38	−0.27	2.21	−0.45	0.00
China	9.90	−0.12	−0.95	7.55	−21.91	14.79	0.00	9.17	−0.44	2.17	−0.44	0.07
Europe	4.70	−0.57	−16.49	12.56	−12.90	11.54	0.00	8.26	0.14	2.70	−0.55	0.00
Rest of World	24.48	−0.26	−6.08	13.38	−7.03	13.50	0.00	7.11	0.55	4.36	−0.89	−0.16
Revenue by destination R_d												
U.S. & CA	−4.97	−0.32	−20.03	17.16	−16.34	4.23	0.00	11.11	−0.84	0.00	0.00	0.05
Mexico	6.54	−0.07	−7.11	9.98	−7.58	2.12	0.00	2.63	1.27	6.99	−1.43	−0.27
Rest of World	−7.91	0.00	−26.75	18.34	−16.36	5.24	0.00	12.22	−0.64	0.00	0.00	0.05

Notes: Both years are in 2023 pesos. In the real-income row the model's price index is held at its E_A value. $\hat{a}, \hat{b}$ are the re-estimated primitives of the continuing firms, measured net of the aggregates. Total $= X_B - X_A$ and Gap $=$ Total $- \sum_k C_k$. Compliance is in percentage points, all other rows in percent of the E_A level.

6.2 The Price of Non-Compliance: Tariff Variations

Figure 4 speaks to the events of March 2025 and February 2026, when the U.S. imposed an in addition to the existing tariffs a 25% surcharge on all non-USMCA compliant exports

[10] It is important to keep in mind how these primitives are obtained in order to read the sign of this column. I back them out in each year taking the aggregates of that year as given. In the case of market reach, for example, $b_{f,d}^{\sigma-1} = R_{f,d}/(p_{f,d}^{1-\sigma}\Phi_d)$, which means that the 2023 primitives are simply what remains of a firm's 2023 outcomes after the 2023 aggregates have been accounted for. Given that the aggregates grew over the period, the residual that is left for the firms is smaller. To state an example, foreign demand on its own would have raised U.S. & CA-bound revenue by 11.11% (column D), but that revenue actually fell by 4.97%, so the re-estimated market reach of the continuing firms has to come out lower in 2023 than in 2018. When these smaller primitives are switched in by themselves, with everything else at its 2018 value, sales and income fall by more or less what the larger aggregates raise them. In this sense the column should be read as the residual relative to the exogenous growth and not as evidence that the firms themselves became less able. Tables 9 and 18 show the direction of the changes, as the mean sourcing efficiency rose in every block between 2018 and 2023 whereas the mean market reach fell in the U.S. & CA and Rest of World markets.

coming in from Mexico. Later on, the U.S. Supreme Court reverse these tariffs and a 10% surcharge in its place. While it is uncertain whether these tariffs will remain in place, they provide an interesting snapshot of a setting when the cost of not complying with the free trade agreement increases dramatically.

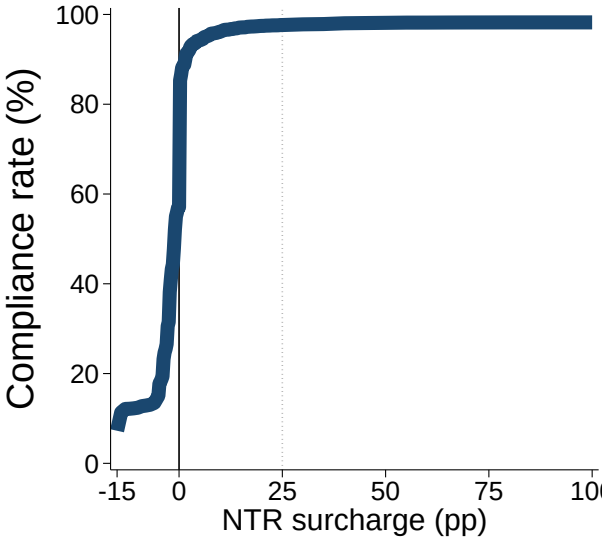
The first result to point is the sharp increase in compliance rates as the tariff increases. While this increase may seem incredibly steep, it speaks to the sharp increase on compliance rates reported as of March 2025 as was seen in Figure 1. In order to avoid the higher tariffs, firms respond by meeting the rules of origin requirements determined in USMCA and source a greater share of their inputs from North American sources and divert away from all other sources outside the free trade region. This turns to be a more costly endeavor for the firm. At the +0% baseline firms were evaluating their most profitable scenario. The profitability of the baseline setting quickly changes as the non-compliance alternative. Buying more North American turns out to be more profitable than paying the new higher tariff. However, the welfare costs are modest, -0.24% under an additional 10% tariff and -0.31% on a 25% increase. This highlights the value of holding a free trade agreement, while meeting rules of origin may be a costly endeavor, it protects the firm against possible high costs of exporting.

6.3 Eliminating the Agreement

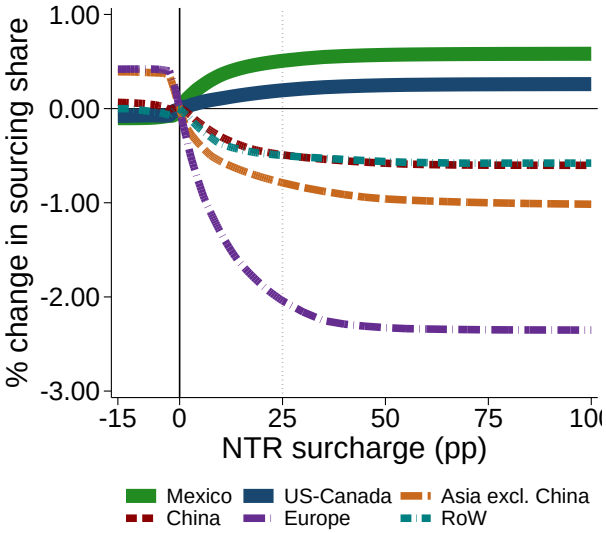
The previous exercise varied the tariff with the agreement in place. Here I remove the agreement and show how eliminating the rule of origin route for firms turns to be more damaging as the cost of not complying increases. This scenario poses a real threat as USMCA is under annual reviews. The agreement can be extended at any moment the three parties to do so, until then reviews are to be held annual until 2036. If no extension is agreed upon, the free trade agreement will expire in July 1, 2036. Once the preferential route is gone, every MX→U.S. & CA shipment has to pay the NTR tariff of its product regardless of its content, meaning that the compliance choice d_{fp} simply disappears. I run this experiment on the export side. In the case of Mexican imports of U.S. and Canadian inputs, these keep their current treatment. Table 14 shows the results for the 2023 calibration.

At the 2023 tariffs, removing the agreement lowers Mexican real income by 1.51%. The value of the agreement increases sharply as tariffs increase. Under the 25% surcharge, the loss of real income increases roughly sixfold to -8.85%. The loss from eliminating the agreement is not concentrated in a few firms but spread over the whole firm block. Under the baseline scenario manufacturing sales fall 4.3% and intermediate purchases fall from every single source by 3.4 to 6.1%, including Mexico's own suppliers. It comes to no surprise that inputs from the U.S. and Canada are the ones that fall the most, given that the firms that lose preferential access are precisely the ones that rely on them the most. The composition of

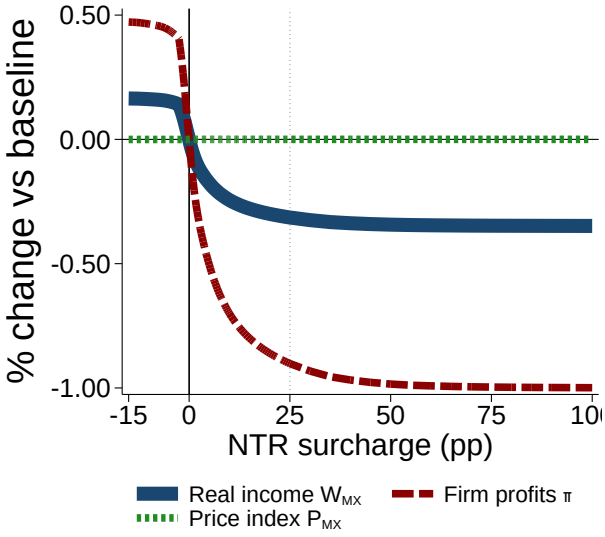
Figure 4: NTR surcharge variation



(a) Compliance



(b) Sourcing composition (shares)



(c) Welfare and components

Notes: Results for 2023. On the horizontal axis is the surcharge (in pp) on top of each product's own NTR rate, with the total floored at zero. 0 is the observed schedule (solid line) and +25 the March 2025 design (dotted line). Panel (a) plots the compliance rate of Section 4.2 (positive-NTR lines). Panel (b) plots the % change in the share of each origin in Mexican sourcing. In panel (c) the price index moves by less than 0.001% along the variation and its line is therefore on top of zero. All changes are relative to the observed schedule.

Table 14: Eliminating the agreement

NTR variation (pp)	−5	Baseline	+5	+25	+100
Aggregate outcomes					
Manufacturing profits Π_M	−1.91	−3.82	−8.68	−22.47	−39.02
Price index P_M	+0.00	+0.00	+0.00	+0.00	+0.01
Real income W_M	−0.76	−1.51	−3.43	−8.85	−15.36
Revenue by destination					
U.S. & CA	−4.10	−8.21	−18.78	−48.81	−84.87
Mexico	−0.76	−1.51	−3.43	−8.85	−15.36
Rest of World	−0.00	−0.00	−0.00	−0.00	−0.00
Intermediate purchases by source					
U.S. & CA	−3.36	−6.15	−11.90	−28.19	−47.78
Mexico	−1.79	−3.36	−7.48	−19.10	−32.93
Asia excl. China	−1.36	−3.42	−10.19	−29.88	−53.64
China	−1.70	−4.38	−11.58	−31.92	−56.45
Europe	−1.64	−3.50	−8.49	−23.99	−43.77
Rest of World	−1.81	−3.66	−8.11	−21.29	−37.46

Notes: The surcharge s is in pp added to each product's own NTR rate, 0 the observed 2023.

sourcing, on the other hand, hardly changes, meaning that the loss is a scale effect on U.S. & CA-bound sales rather than a reallocation. When compared to the content-rule experiments the loss is large, and the reason behind this is how much of the economy is exposed to it. U.S. & CA-bound sales make up 42% of the revenue of the panel (Section 5.1)There is also the question of how a loss in profits translates into real income. Manufacturing profits Π_M, counting traders and non-traders, are roughly two fifths of the income measure, and the wage bill at the pinned wage is the remaining three fifths. About two fifths of the proportional loss in profits therefore shows up in real income.

A natural question that may arise is if firms would substitute towards other destinations in the case where access to the U.S. & Canada markets becomes prohibitively expensive. Total manufacturing sales to this destination accounts for roughly half of manufacturing revenue once the domestic market is accounted for. The model does not allow for this kind of substitution. Markets are separable, since marginal cost is constant and common to all destinations and foreign demand is held fixed, meaning that what a firm sells in the Rest of World does not depend on what happens in the U.S. & CA market. The extensive margin does not change this either. A firm enters the Rest of World when its market reach and the fixed cost of entry justify it, and neither of the two moves when the agreement goes away, which is why revenue in the Rest of World does not change. Revenue from Mexico does fall, this result arises from the endogenous outcome in local aggregate income. As manufacturing profits fall from the decline in sales from the other North American countries, domestic

income falls with it. This drags the domestic market to shrink along along with them.

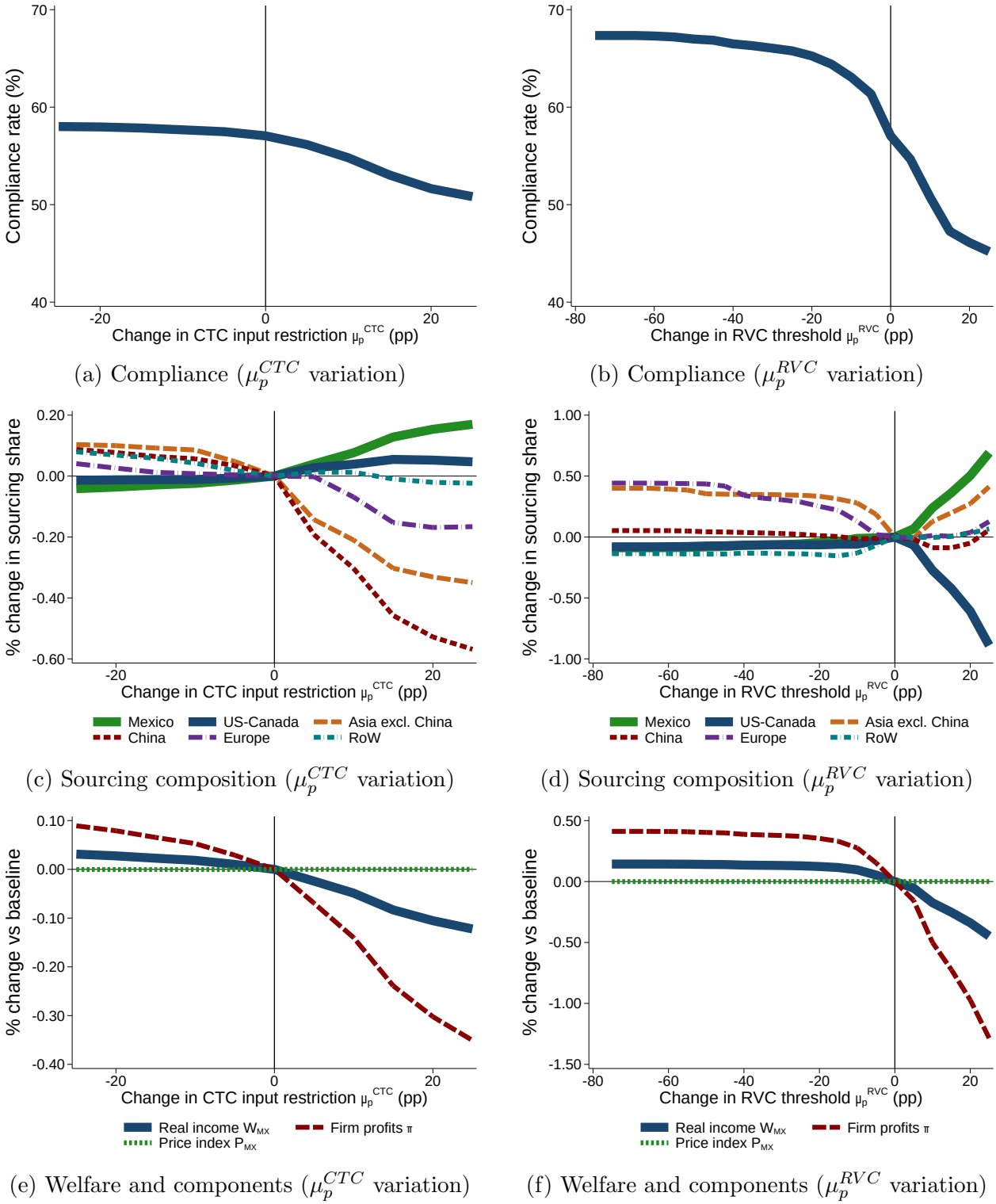
The comparison with Section 6.2 shows where this value comes from. There, with the preferential route available, the same surcharges cost at most a third of a percent of real income, because firms comply instead of paying. Without the route there is nothing to switch to. What the agreement provides is therefore protection against the tariff, and that protection is worth more the larger the tariff is. The jump in compliance after the 2025 order (Section 2) is this protection being used by the firms. All of this value runs through the preferential route, which is why the next exercise moves the content rules that the route depends on.

6.4 Rules-of-Origin Strictness

Section 6.1 found that the 2020 switch in rules cost little, mostly because the outside tariff was low. Here I tighten the rules directly. I do it at observed tariffs first, and in the second half of the section with variations on tariff surcharges on top. Figure 5 varies the two content parameters around their USMCA baselines, holding everything else fixed. The CTC input restriction μ_p^{CTC} goes over $\pm$25pp and the RVC threshold μ_p^{RVC} over $[-75, +25]$pp. Compliance moves a lot under both, but differently by regime.

I take two things from this figure. First, compliance responds considerably to the strictness of the agreement. Moving μ_p^{CTC} by 25pp in either direction from the USMCA baseline changes compliance by $+1.0$pp when the restriction is loosened and by -6.2pp when it is tightened, so 7pp from one end to the other. The μ_p^{RVC} variation over $[-75, +25]$pp gives a wider 22pp range ($+10.3$pp at the loose end and -11.9pp at the tight end). Since, under the lens of the model, compliance is a gradual decision that the firm can adjust for by buying more or less North American content, the response on compliance shows a smooth panel. These adjustments in compliance are adjustest through the sourcing compositions that firms decide upon. Lower content threshold requirements give firms more relaxed constraints on comlpying and thus opt to soruce more freely from other sources. Higher content threshold show an ambiguous results. While higher CTC requirements unilaterally push for more North American sourcing in both Mexico and the U.S. & Canada bundle, higher RVC requirements push for less U.S. & Canada sourcing. While this may seem counterintuitive at first, it shows the results coming from the sharper deceases in compliance. In this case, higher content requirements simply push for a generalized preference to pay tariff instead. Note that the response is regime-specific by construction, as only CTC and Hybrid move under the μ_p^{CTC} variation, whereas only RVC and Hybrid move under the μ_p^{RVC} variation. Meaning that the degree of responsivness betweeen the two compliance regimes may be attributed to the specific firm composition and product-level rules.

Figure 5: Counterfactual variations of rules-of-origin strictness



(a) Compliance (μ_p^{CTC} variation)

(b) Compliance (μ_p^{RVC} variation)

(c) Sourcing composition (μ_p^{CTC} variation)

(d) Sourcing composition (μ_p^{RVC} variation)

(e) Welfare and components (μ_p^{CTC} variation)

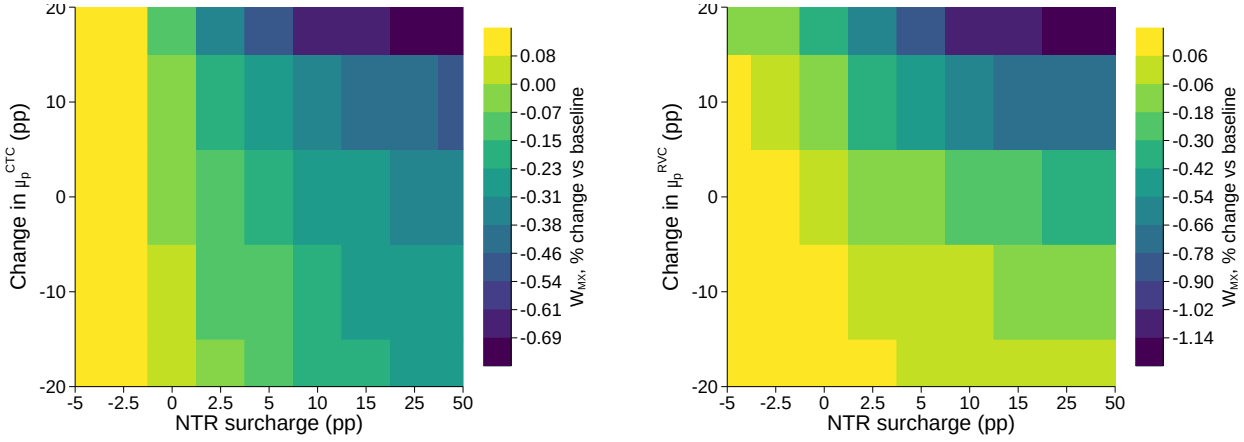
(f) Welfare and components (μ_p^{RVC} variation)

Notes: Results for 2023. Horizontal axis: change in pp from the USMCA value of μ_p^{CTC} (left column) or μ_p^{RVC} (right column); the vertical line marks the baseline. Top row: compliance rate of Section 4.2 (positive-NTR lines). Middle row: % change relative to the baseline in each origin's share of Mexican sourcing, $\bar{\mu}_s = M_{MX,s}/\sum_{s'} M_{MX,s'}$. Bottom row: % change in real income, the traders' profits (non-traders excluded) and the Mexican price index. Given that the price index moves by less than 0.001% along these variations, its line coincides with zero.

The second takeaway is that the welfare panels (e, f) are not flat. Tightening μ_p^{CTC} by 25pp lowers W_M by 0.12% and the traders' profits by 0.35%, whereas tightening μ_p^{RVC} by 25pp lowers them by 0.45% and 1.28%, and loosening raises W_M by up to 0.03% and 0.14% respectively. The price index moves by less than a thousandth of a percent and the sourcing shares by at most 0.9% (panels c, d), meaning that this cost works through profits rather than through prices or through the composition of sourcing. The intuition behind this is the following. When the rule tightens most firms remain compliant, as compliance is still 45 to 51% of U.S. & CA-bound revenue at the tight ends, and in order to do so they buy North American content that they would otherwise not buy. Firms are willing to pay this cost because preference is valuable, given that at observed tariffs the agreement is worth 1.51% of real income (Section 6.3). In other words, tightening the RVC threshold by 25 points costs about three tenths of what the agreement is worth, and the second half of this section shows that this cost becomes larger as non-compliance becomes more expensive.

Whereas the variations above and the surcharge grid of Section 6.2 move one instrument at a time, Figure 6 crosses them. What I do is to vary each content parameter over ±20pp against a nine-point grid of NTR surcharges, which goes from −5pp (close to free trade), through 0 (the observed schedule), up to +50pp, where +25pp corresponds to the 2025 design. Each panel then shows the real-income surface that comes out of this.

Figure 6: Rules-of-origin strictness at different levels of the outside tariff



(a) Real income (μ_p^{CTC} variation) (b) Real income (μ_p^{RVC} variation)

Notes: Results for 2023. Each cell corresponds to one (content parameter, NTR surcharge) pair and is shaded by the % change in real income relative to the observed schedule at baseline rules. Colour goes from dark for low to bright for high on a viridis ramp, and each panel has its own scale, shown in its legend. The surcharge steps are unevenly spaced but drawn at equal width, meaning that the horizontal axis is ordinal and not to scale.

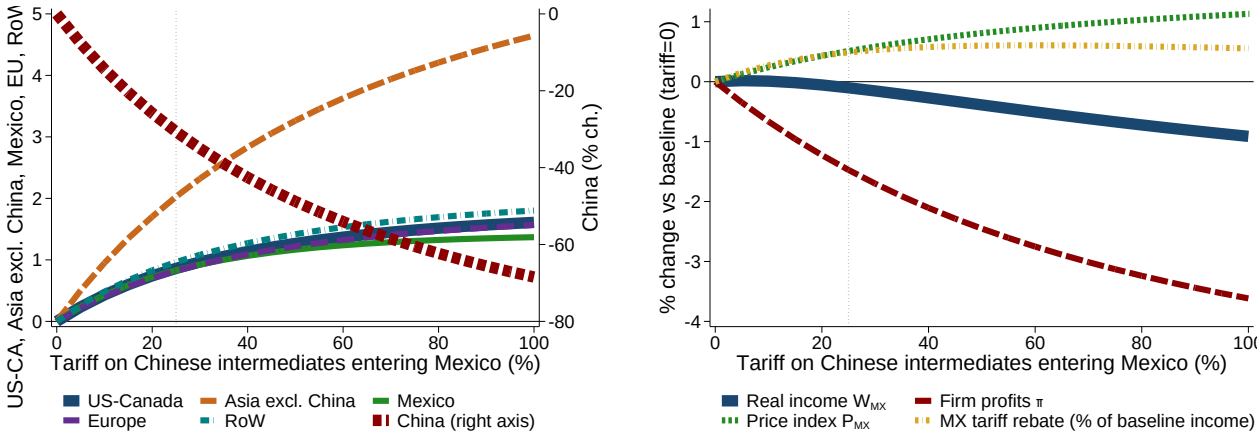
What both panels show is that content rules matter more when not complying is expensive. At observed tariffs, tightening μ_p^{CTC} or μ_p^{RVC} by 20 points costs 0.11% and 0.34%

of real income, whereas at the 25-point surcharge of 2025 the same tightening costs 0.42% and 0.90%, given that firms remain compliant and buy North American content rather than paying the tariff. Given that content rules work on the North American share of inputs, the last exercise targets the Chinese share directly instead.

6.5 Tariffs on Chinese Intermediates

The last experiment is a stylized version of the tariffs that Mexico imposed in January 2026, which raised duties to 20 to 35% (up to 50%) on 1,463 tariff lines from countries without a trade agreement, China included. I model it as a uniform tariff τ_{CH} on Chinese intermediates entering Mexico, which is varied from 0 to 100% (Figure 7). Given that $\psi_s \equiv T_s(c_s\tau_s)^{-\theta}$, raising the effective input price of Chinese intermediates $w_{CH}\tau_{CH}$ by the factor $(1+\tau_{CH})$ is equivalent to scaling the sourcing shifter by $(1+\tau_{CH})^{-\theta}$, so the counterfactual enters the model as $\psi_{CH} \to \psi_{CH} \cdot (1+\tau_{CH})^{-\theta}$. The tariff revenue is rebated lump-sum to the representative Mexican consumer, assuming a balanced budget, and enters the numerator of (19) as a fiscal transfer, given that there is no tariff-revenue term there at baseline.

Figure 7: Tariffs on Chinese intermediates entering Mexico



(a) Sourcing reallocation

(b) Welfare and components

Notes: Results for 2023. Panel (a): % change in imports by origin (China on right axis), tariff-inclusive spending; dotted line at 25%.

Panel (a) shows the sourcing response. A 25% tariff on Chinese intermediates cuts Chinese sourcing by 31%, at 50% the cut goes up to 49%, and it takes a 100% tariff to reach 68%. Note that these are tariff-inclusive purchases, whereas at customs value the cuts are 45, 66 and 84%. The volume that is displaced is small in aggregate, and the source that gains the most in proportion is outside North America, as at 25% Asia excluding China rises by 2.0%, whereas Europe, the Rest of World and U.S. & CA rise by 0.8 to 1.0% and Mexican domestic

sourcing by 0.8%. This is in line with Asia excluding China being the block whose sourcing efficiencies co-move the most with those of China (Table 12a), which suggests to me that the policy first pushes firms toward geographies whose input cost and productivity draws look like those of China, and that this substitution pattern ends up limiting how much trade it is able to redirect specifically to North America.

Panel (b) shows the welfare implications of the same policy. Small tariffs leave real income essentially unchanged, at +0.02% for a 5% tariff and +0.01% at 10%, given that the tariff rebate roughly offsets the fall in profits and the rise in the price index. At 25, 50 and 100%, the change in W_M is −0.10, −0.39 and −0.91%, with the price index rising by 0.5, 0.8 and 1.1% and the traders' profits falling by 1.5, 2.5 and 3.6%, net of a tariff rebate worth 0.5 to 0.6% of income. Aggregate compliance rises slightly across the variation (from 57.1% at zero tariff to 57.4% at 25% and 58.0% at 100%), as forcing firms off Chinese inputs raises their North American content and pushes a small mass of them over the compliance threshold. When compared to the other instruments, a 25% upstream tariff costs a tenth of a percent of real income while cutting Chinese sourcing by far more than any content-rule variation, which leaves sourcing shares within one percent of baseline, whereas a 25-point tightening of the RVC threshold costs 0.45% of real income and eliminating the agreement costs 1.51% (Section 6.3). Thus, if the policy goal is to have less Chinese content in North American supply chains, upstream input tariffs work at a margin where rules of origin are essentially inert. The trade-off is that they raise sourcing from the rest of Asia proportionally more than from North America. It is worth pointing out that this is also the policy that pushes against the shift toward Chinese inputs which the decomposition of Section 6.1 attributed to sourcing conditions.

7 Conclusion

This paper studies how firms organize North American supply chains when tariff-free access to the U.S. market must be reached with rules-of-origin compliance, and what this trade-off implies for Mexican welfare. In particular, the model allows one to quantify how much the protection that USMCA provides against U.S. tariff threats is worth to Mexico. When the tariff on non-compliant goods goes up, firms respond by complying with the product-specific content rules instead of paying it, which allows them to avoid most of the cost of the tariff. The value of holding the agreement therefore grows with the size of the threat. At the NTR tariffs in place before 2025, eliminating USMCA would cost Mexico 1.51% of its real income, whereas at the 25-point surcharge of the 2025 order the loss would reach 8.85%, about six times as much. With the agreement in place, on the other hand, the same surcharge costs

only 0.31%. Through the lens of the model, this is what prevented the 2025 tariffs from turning into a significant economic downturn for Mexico.

Without additional tariffs, on the other hand, staying outside of the agreement was cheap. NTR rates on Mexican manufactures are low, and many firms chose to pay them and keep sourcing freely. Having had a low cost of not complying with NAFTA or USMCA allows to understand the fall in preference ultilization when the more stringent USMCA was introduced. This result us explains why the move from NAFTA to USMCA rules is close to welfare-neutral in the model, since a firm for which the new rules were binding could just pay the tariff. Across the two periods, the part of the aggregate income change that I am able to interpret comes from foreign-demand growth and from sourcing conditions shifting toward Chinese inputs. Furthermore one can rationalize the sharp jump in compliance that only came when the 2025 surcharge made non-compliance costly Whatever the rule type, compliance reacts strongly to what it costs not to comply, and the welfare cost of the rules rises with that cost.

The policy implications come from the same place. The cost of content rules depends on the tariff environment. Even at historical tariffs, tighter rules cost real income, since firms keep complying and pay for North American content rather than give up preference. Raising the regional value content (RVC) threshold by 25 points costs 0.45% of real income, roughly three tenths of the value of the agreement. The same 25 points on the change-in-tariff-classification (CTC) input restriction cost 0.12%. Put the 2025 tariff in place and these costs grow three- to fourfold, to more than what the tariff itself costs. I read this as rules of origin and the outside tariff being complements, because the tariff decides whether firms comply and therefore how much the rules cost. Upstream tariffs on Chinese intermediates are different. At a 25% rate they reallocate sourcing for about a tenth of a percent of real income. However, if the goal is to push for more North American content, this may not be the case as the rest of Asia gains proportionally more than North America does. The reason falls in capabilities of the firm per-se. Those who are more capable of sourcing from China, tend to be more capable of sourcing from other Asian suppliers as well.

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A Model Derivations

This appendix goes through the derivations behind the closed-form expressions of Section 3. I begin with the Fréchet closed forms for the intermediate cost index and the sourcing shares (Section A.1) and with the Cobb-Douglas unit cost (Section A.2). Sections A.3 and A.4 derive the compliant unit cost under CTC and the cheapest compliant restriction under RVC. Section A.5 explains how I solve the firm problem. Section A.6 derives the Mexican CES price index that I use in the closure of the Mexican block, Section A.7 decomposes manufacturing labor demand and, lastly, Section A.8 goes over the domestic-demand loop.

A.1 Fréchet Closed Forms

Conditional on the firm's sourcing set $\mathcal{S}_f$, the cost of sourcing input j from country s is given by

$$c_{f,s}(j) = \frac{c_s \tau_s}{z_s(j)},$$

where $z_s(j) \sim \text{Fréchet}(T_s a_{f,s}^{\theta}, \theta)$ is an idiosyncratic productivity draw, independent across countries and inputs. The firm picks the lowest-cost source for each input,

$$s^*(j) = \arg\min_{s \in \mathcal{S}_f} c_{f,s}(j).$$

Standard Fréchet algebra (Eaton and Kortum, 2002) gives the probability that country s offers the minimum cost as

$$\mu_{f,s}(\mathcal{S}_f) = \int_0^1 \mathbb{1}\{s^*(j) = s\}\, dj = \frac{T_s a_{f,s}^{\theta} (c_s \tau_s)^{-\theta}}{\sum_{s' \in \mathcal{S}_f} T_{s'} a_{f,s'}^{\theta} (c_{s'} \tau_{s'})^{-\theta}}, \tag{29}$$

which is equation (7) of the main text. The intermediate cost index then comes from integrating the minimum cost over the Fréchet distribution,

$$c_f^M(\mathcal{S}_f) = \Gamma\left(1 - \frac{1}{\theta}\right)\left(\sum_{s\in\mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}\right)^{-1/\theta}, \tag{30}$$

that is, equation (6). A country with a higher technological capability T_s, a higher firm-country sourcing efficiency $a_{f,s}$ or a lower effective cost $c_s\tau_s$ takes a larger share of the firm's intermediate sourcing.

A.2 Cobb-Douglas Unit Cost

The firm's production function combines Mexican labor ℓ and an intermediate composite M through a constant-returns Cobb-Douglas:

$$q_f = \ell_f^{\gamma} M_f^{1-\gamma}, \qquad \gamma \in (0,1).$$

Taking the Mexican wage w_m and the price of the intermediate composite $c_f^M(\mathcal{S}_f)$ from (6) as given, the firm picks ℓ and M so as to produce q_f units at the lowest cost:

$$\min_{\ell,M}\; w_m\,\ell + c_f^M\,M \quad \text{s.t.} \quad \ell^{\gamma}M^{1-\gamma} \geq q_f.$$

The Lagrangian is $\mathcal{L} = w_m\ell + c_f^M M - \lambda(\ell^{\gamma}M^{1-\gamma} - q_f)$, with first-order conditions

$$w_m = \lambda\gamma\,\ell^{\gamma-1}M^{1-\gamma}, \qquad c_f^M = \lambda(1-\gamma)\,\ell^{\gamma}M^{-\gamma}.$$

From the ratio of the two FOCs one gets the cost-share property of Cobb-Douglas,

$$\frac{w_m\,\ell}{c_f^M\,M} = \frac{\gamma}{1-\gamma}, \tag{31}$$

i.e. labor's share of total variable cost is γ, the exponent in the production function. I rely on this identity for the calibration $\hat{\gamma} = wL/(wL + M_t)$ of Section 4.

From (31) one can write labor at the optimum as a function of the intermediate composite,

$$\ell = \frac{\gamma}{1-\gamma}\,\frac{c_f^M}{w_m}\,M.$$

Substituting this into the binding production constraint $\ell^{\gamma}M^{1-\gamma} = q_f$ leaves a single equation

in M:

$$q_f \;=\; \left(\frac{\gamma}{1-\gamma}\,\frac{c_f^M}{w_m}\,M\right)^{\gamma} M^{1-\gamma} \;=\; \left(\frac{\gamma}{1-\gamma}\right)^{\gamma}\left(\frac{c_f^M}{w_m}\right)^{\gamma} M,$$

where the second equality uses $M^{\gamma}\cdot M^{1-\gamma}=M$. Solving for M and going back to the labor expression, the conditional input demands are

$$\ell_f^*(q_f) = q_f\left(\frac{\gamma}{1-\gamma}\,\frac{c_f^M}{w_m}\right)^{1-\gamma}, \qquad M_f^*(q_f) = q_f\left(\frac{1-\gamma}{\gamma}\,\frac{w_m}{c_f^M}\right)^{\gamma}.$$

Plugging these into the cost objective $w_m\ell_f^* + c_f^M M_f^*$ gives total cost

$$C(q_f) = q_f\, w_m^{\gamma}\,(c_f^M)^{1-\gamma}\left[\left(\frac{\gamma}{1-\gamma}\right)^{1-\gamma} + \left(\frac{1-\gamma}{\gamma}\right)^{\gamma}\right] = q_f\cdot c_f.$$

The term in brackets is a constant. Factoring out $\gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)}$, the first summand becomes γ and the second $(1-\gamma)$, and the bracket is then $\gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)}(\gamma+(1-\gamma)) = \gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)} \equiv B$. The unit cost is therefore

$$c_f(\mathcal{S}_f) = B\, w_m^{\gamma}\left[c_f^M(\mathcal{S}_f)\right]^{1-\gamma}, \qquad B \equiv \gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)}, \tag{32}$$

which is equation (9) of the main text. It is worth pointing out that the constant B has no economic meaning, as it is simply what is left over from the Cobb-Douglas algebra.

A.3 CTC Compliant Unit Cost

Under CTC compliance a subset of measure μ_p^{CTC} of the firm's continuum of input varieties $j \in [0,1]$ has to come from North America. Input varieties are drawn iid Fréchet across countries. With a continuum of them, the law of large numbers implies that a subset of measure μ_p^{CTC} contributes to cost according to its measure, whichever varieties end up in it. I can therefore split the input continuum into a restricted and an unrestricted sub-continuum and give each its own effective Fréchet scale. For the restricted, NA-only subset the effective scale is $\sum_{s\in NA\cap\mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}$ and for the unrestricted subset it is $\sum_{s\in\mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}$. Combining the two with Cobb-Douglas weights then gives the compliant intermediate cost index:

$$c_{f,p}^{M,CTC}(\mathcal{S}_f) = \Gamma\!\left(1-\frac{1}{\theta}\right)\left(\sum_{s\in NA\cap\mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}\right)^{-\mu_p^{CTC}/\theta}\left(\sum_{s\in\mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}\right)^{-(1-\mu_p^{CTC})/\theta}.$$

Substituting this into the Cobb-Douglas structure $c_f = B\, w_m^{\gamma}[c_f^M]^{1-\gamma}$, the compliant unit cost is:

$$c_{f,p}^{CTC}(\mathcal{S}_f) = B\, w_m^{\gamma}\left[\Gamma\big(1-\tfrac{1}{\theta}\big)\right]^{1-\gamma} \times \left(\sum_{s\in NA\cap\mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}\right)^{-\mu_p^{CTC}(1-\gamma)/\theta} \left(\sum_{s\in\mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}\right)^{-(1-\mu_p^{CTC})(1-\gamma)/\theta}. \quad (33)$$

The exponents on the intermediate terms carry a $(1-\gamma)$ because under Cobb-Douglas production only the $(1-\gamma)$ share of total cost goes to intermediates. If I write $\mu_f^{NA,M} \equiv \sum_{s\in NA\cap\mathcal{S}_f} \mu_{f,s}(\mathcal{S}_f)$ for the North American share of the unrestricted intermediate bundle, (33) reduces to the compact form that the solver uses,

$$c_{f,p}^{CTC}(\mathcal{S}_f) = c_f(\mathcal{S}_f)\left(\mu_f^{NA,M}\right)^{-\mu_p^{CTC}(1-\gamma)/\theta},$$

since $\sum_{s\in NA\cap\mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta} = \mu_f^{NA,M}\sum_{s\in\mathcal{S}_f} T_s a_{f,s}^{\theta}(c_s\tau_s)^{-\theta}$ by (7).

A.4 RVC Compliant Unit Cost

Under an RVC threshold it is the firm that chooses the restricted measure. If a measure α of the input continuum is restricted to North American sources, the unrestricted sub-continuum keeps its NA share $\mu_f^{NA,M}$, the NA share of the whole intermediate bundle is $\alpha+(1-\alpha)\mu_f^{NA,M}$ and the content share of production is $\gamma+(1-\gamma)\big[\alpha+(1-\alpha)\mu_f^{NA,M}\big]$. Setting the latter equal to μ_p^{RVC} and solving for α,

$$\alpha_{f,p}^{*}(\mathcal{S}_f) = \frac{\mu_p^{RVC}-\gamma-(1-\gamma)\mu_f^{NA,M}}{(1-\gamma)\big(1-\mu_f^{NA,M}\big)} = \frac{\mu_p^{RVC}-\mu_f^{NA}(\mathcal{S}_f)}{(1-\gamma)\big(1-\mu_f^{NA,M}\big)},$$

clamped to $[0,1]$, so a firm that is already above the threshold restricts nothing. The restricted cost increases with the restricted measure, which makes α^* the cheapest compliant restriction, and the RVC compliant cost is $c_{f,p}^{RVC}(\mathcal{S}_f) = c_f(\mathcal{S}_f)\,(\mu_f^{NA,M})^{-\alpha_{f,p}^{*}(1-\gamma)/\theta}$. A firm without a North American source has $\mu_f^{NA,M}=0$ and cannot comply whenever $\alpha^*>0$.

A.5 Solving the Firm Problem

Once the sourcing set is fixed, profits in (14) are separable across products and across destinations. Because of this, solving the firm problem only requires enumerating sourcing sets, which I do in three steps.

First, for each of the $2^{|\mathcal{S}|}-1$ non-empty sourcing sets $\mathcal{S}_f \subseteq \mathcal{S}$ I compute the unrestricted unit cost $c_f(\mathcal{S}_f)$ from (9) and the North American share $\mu_f^{NA}(\mathcal{S}_f)$.

Second, given $\mathcal{S}_f$ the compliance choice d_{fp} only affects the U.S. & CA revenue of product p, so the firm simply takes the cheaper of the two routes product by product,

$$c_{f,p,\text{U.S. \& CA}}(\mathcal{S}_f) \;=\; \min\Big\{c_{f,p}(\mathcal{S}_f),\; (1+\tau_p^{NTR})^{\sigma/(\sigma-1)}\, c_f(\mathcal{S}_f)\Big\},$$

with $d_{fp}=1$ whenever the compliant cost is weakly lower.

Third, given $\mathcal{S}_f$ and the compliance choices, destinations enter profits additively. The firm serves d if and only if $\frac{1}{\sigma}\sum_{p\in\mathcal{P}_f} R_{f,p,d} > w_m\kappa_d$, with $R_{f,p,d}$ from (13), and this condition replaces a search over the eight destination sets.

A.6 Mexican CES Price Index

The CES price index P_M of equation (18) is the usual demand-side dual of the destination-d aggregator in (3). Starting from

$$\max_{\{q_d(\omega)\}} \left(\int_{\omega\in\Omega_d} \big(b_d(\omega)q_d(\omega)\big)^{(\sigma-1)/\sigma} d\omega\right)^{\sigma/(\sigma-1)} \quad \text{s.t.} \quad \int p_d(\omega)q_d(\omega)d\omega = E_d,$$

the first-order condition at ω gives

$$b_d(\omega)^{(\sigma-1)/\sigma}\, q_d(\omega)^{-1/\sigma} \;=\; \lambda\, p_d(\omega),$$

which can be inverted for the demand function:

$$q_d(\omega) \;=\; \lambda^{-\sigma}\, b_d(\omega)^{\sigma-1}\, p_d(\omega)^{-\sigma}.$$

Plugging this back into the budget constraint pins down the multiplier:

$$E_d \;=\; \lambda^{-\sigma}\int b_d(\omega)^{\sigma-1}\, p_d(\omega)^{1-\sigma} d\omega \;\Rightarrow\; \lambda^{-\sigma} \;=\; \frac{E_d}{\int (p_d(\omega)/b_d(\omega))^{1-\sigma} d\omega}.$$

If the CES price index P_d is defined implicitly by $\lambda^{-\sigma} = E_d P_d^{\sigma-1}$, so as to get back the demand expression (4), $q_d(\omega) = E_d P_d^{\sigma-1} p_d(\omega)^{-\sigma} b_d(\omega)^{\sigma-1}$, then the price index satisfies

$$P_d^{1-\sigma} \;=\; \int_{\omega\in\Omega_d} \left(\frac{p_d(\omega)}{b_d(\omega)}\right)^{1-\sigma} d\omega. \tag{34}$$

In the model varieties are firm-product pairs $\{(f,p)\}$, not a continuum. The active

varieties sold at destination d in equilibrium are $\{(f,p) : f \in \mathcal{A}, d \in \mathcal{D}_f, p \in \mathcal{P}_f\}$, the appeal weight is the firm-level market reach $b_{f,d}$, which all of the firm's products share at destination d, and the price is $p_{f,p,d}$. Replacing the integral in (34) by the corresponding sum and setting $d = MX$ gives equation (18):

$$\left(P_M\right)^{1-\sigma} = \int_{f\in\mathcal{A}:\, MX\in\mathcal{D}_f} \sum_{p\in\mathcal{P}_f} \left(\frac{p_{f,p,MX}}{b_{f,MX}}\right)^{1-\sigma} df.$$

The $(1-\sigma)$ exponents on the two sides cannot be cancelled against each other, since the right-hand side is a sum of $(p/b)^{1-\sigma}$ terms. One can also write it in rooted form, $P_M = \left(\sum (p/b)^{1-\sigma}\right)^{1/(1-\sigma)}$.

A.7 Manufacturing Labor Demand Decomposition

I split total Mexican manufacturing labor compensation $w_m L_m$ over the active set $\mathcal{A}$ into three parts, and production labor is the first. With Cobb-Douglas production, a share γ of each firm's variable cost, $VC_f \equiv \frac{\sigma-1}{\sigma}\sum_{d\in\mathcal{D}_f}\sum_{p\in\mathcal{P}_f} R_{f,p,d}$, is paid straight to Mexican production workers. Summing over firms,

$$E^{\text{prod}} = \int_{f\in\mathcal{A}} \gamma \cdot VC_f \, df.$$

Second, there is the $(1-\gamma)\mu_{f,MX}$ share of variable cost that goes to intermediates bought in Mexico. I assume that upstream Mexican suppliers earn no economic profit and only use labor, which makes this spending Mexican labor compensation as well, just one stage further up the chain:

$$E^{\text{int}} = \int_{f\in\mathcal{A}} (1-\gamma)\, \mu_{f,MX}(\mathcal{S}_f) \cdot VC_f \, df.$$

The last part is the fixed costs. Each active firm pays $w_m \sum_{s\in\mathcal{S}_f} \kappa_s$ in sourcing fixed costs and $w_m \sum_{d\in\mathcal{D}_f} \kappa_d$ in destination entry costs, both in units of Mexican labor by assumption, which adds

$$E^{\text{fixed}} = \int_{f\in\mathcal{A}} w_m \Big(\sum_{s\in\mathcal{S}_f} \kappa_s + \sum_{d\in\mathcal{D}_f} \kappa_d\Big) df.$$

The integral only runs over the traders. In the calibration as executed, the non-trading bins are not in the firm block, and they come in through the price-index term K and their domestic sales instead (Section 4). Bringing them back as firm rows would be simple, as they would enter this accounting with $\mathcal{S}_f = \{\text{MX}\}$, $\mathcal{D}_f \subseteq \{\text{MX}\}$ and $\mu_{f,MX} = 1$, and a variable bracket of just VC_f. Adding the three parts gives $w_m L_m$, and dividing by the pinned wage

$w_m = A_o$ gives the labor-demand equation (17) of the main text:

$$w_m L_m = \int_{f\in\mathcal{A}} \Big[\big(\gamma + (1-\gamma)\mu_{f,MX}\big) VC_f + w_m \sum_{s\in\mathcal{S}_f} \kappa_s + w_m \sum_{d\in\mathcal{D}_f} \kappa_d \Big] df.$$

A.8 Solving the Domestic-Demand Loop

Under the Cobb-Douglas upper tier (2), Mexican manufacturing expenditure moves with Mexican income,

$$E_M \;=\; E_M^0 \,\frac{I}{I^0}, \qquad I \equiv w_m \bar{L} + \Pi_M + \Pi_o, \tag{35}$$

where the superscript 0 stands for master-equilibrium values. Mexican manufacturing spending therefore moves at the baseline propensity E_M^0/I^0, which is 0.60 in 2018 and 0.68 in 2023, and not at χ. E_M^0 is the baseline domestic manufacturing sales of the model firms and the non-traders (Section 4). I anchor it on these sales and not on χI^0 because domestic sales are gross of the intermediates that firms sell to each other. What each counterfactual then does is to iterate on (E_M, P_M). For a given pair, firms solve (14) and their choices update Π_M through (20) and P_M through (18), and the loop only stops once (35) and (18) both hold at the choices of the firms. Note that the master equilibrium is, by construction, a fixed point of this loop.

B SCIAN6–HS6 Concordance and Match Quality

I take the CTC input-restriction parameter μ_p^{CTC} from INEGI's 2018 Mexican input–output (I-O) table. That table is organized by six-digit SCIAN industry, while the rules of origin and the model are at HS6, so I need a bridge between the two. On the HS side I use the NAICS6↔HS6 concordance of Pierce and Schott (2012), weighted by MX→US and MX→CA export value. To get from NAICS to SCIAN I use a crosswalk that goes as deep as INEGI reports, which is 6, 5 or 4 digits depending on the industry. Pushing the SCIAN I-O flows through both the output and the input side of this concordance gives an HS6×HS6 input–output matrix, and μ_p^{CTC} is the restricted-input share of each output HS6 in that matrix. Some HS6 codes in the legal list have no I-O backing of their own. For those I use the flow-weighted mean of their HS4 siblings, then of their HS2 siblings, and if neither exists, zero. A code from another HS revision gets the value of whichever codes it links to in the UN conversion tables. There are also a handful of codes whose rule does not restrict any input, and these get a zero from the rule directly.

The translation shows to be successful where trade actually concentrates. Table 15 reports the match rates by treaty, in counts and weighted by MX→U.S. & CA export value. Under

USMCA only 60% of the HS6 codes have μ_p^{CTC} computed straight from the I-O, but those codes carry 91–93% of trade value. What does not match is a long tail of codes with very little trade. Vehicles, electronics and machinery all map cleanly. The revision link accounts for 7% of codes and 7% of 2023 trade value, mostly codes renumbered at the HS2022 boundary. NAFTA has a lower direct-coding share, 49% of codes and 72% of trade value through the Annex 401 parse of Conconi et al., which leaves 13–21% of value to the HS2-level imputation. Zero fallbacks barely matter for either treaty (less than 2% of codes, 0.2% of trade value). In both periods the parameter file covers all MX→U.S. & CA manufacturing (HS2 16–97) export value.

Table 15: SCIAN6–HS6 match quality for the CTC input-restriction parameter μ_p^{CTC}

			% of modeled trade value	
Provenance of μ_p^{CTC}	HS6 codes	% of HS6	2018	2023
Panel A. USMCA				
Direct I-O HS6 match	3,032	59.7	93.1	91.0
HS4 match	875	17.2	0.2	0.3
HS2 match	497	9.8	4.7	0.5
HS revision link	348	6.9	0.4	6.6
Rule (no restricted inputs)	12	0.2	0.0	0.0
Yarn-Forward rail	277	5.5	1.5	1.5
Zero fallback	36	0.7	0.2	0.2
Panel B. NAFTA				
Direct I-O HS6 match	2,482	48.9	72.2	72.1
HS4 match	1,136	22.4	5.2	6.0
HS2 match	901	17.7	21.1	13.3
HS revision link	206	4.1	0.0	7.1
Yarn-Forward rail	258	5.1	1.3	1.3
Zero fallback	94	1.9	0.2	0.2

Notes: The 5,077 manufacturing HS6 (HS2 16–97) of the parameter file, which include codes of the HS2012, HS2017 and HS2022 revisions. Value columns weighted by 2018/2023 MX→U.S. & CA exports (UN Comtrade, Mexico as reporter).

C CE–PEME Match Quality and Diagnostics by Period

The trader panel is built by joining the CE manufacturing firms onto the PEME firm×HS6 customs panel on the firm identifier. Table 16 reports the diagnostics of this match as they

come out of the laboratory.

Table 16: CE–PEME match diagnostics by period

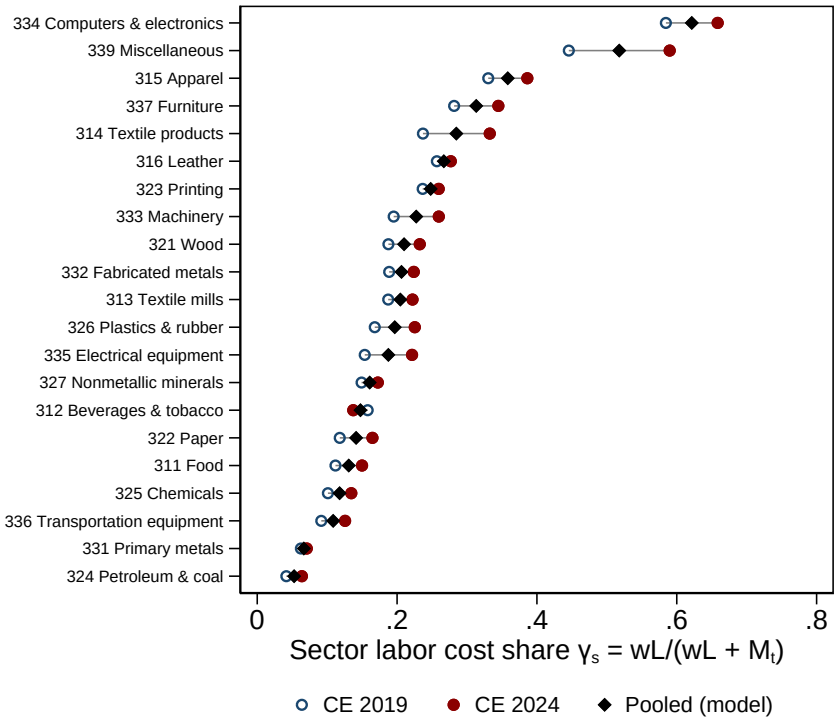
	CE 2019 × PEME 2018	CE 2024 × PEME 2023
PEME manufacturing firms	6,853	6,345
matched to a CE firm	6,677 (97.4%)	5,244 (82.6%)
PEME export value matched	99.0%	86.0%
PEME import value matched	98.9%	83.1%
CE manufacturing exporters ($rev_f > 0$)	5,224	4,781
matched to a PEME firm	4,068 (77.9%)	3,180 (66.5%)
CE foreign revenue matched	96.9%	86.4%
CE manufacturing importers ($M_f > 0$)	3,202	2
matched to a PEME firm	2,359 (73.7%)	2
Trader panel (firms)	5,726	5,148
PEME firm-level trade (USD)		
Exports: mean / median	\$60.4m / \$3.4m	\$84.8m / \$5.8m
Imports: mean / median	\$48.8m / \$3.8m	\$62.6m / \$5.3m
HS6 products per firm: mean / median	27.1 / 9	28.3 / 10
CE foreign revenue / PEME exports, matched firms		
p25	0.0085	0.0104
p50	0.0189	0.0178
p75	0.0252	0.0274
mean	39.8	1.09

Notes: Firms are joined directly on the firm identifier within the manufacturing universe. The ratio divides CE foreign revenue (thousand pesos) by PEME exports (USD), and for reference, exchange-rate parity is ≈ 0.019 in 2018 and 0.018 in 2023.

Out of the matched firms I keep those with positive revenue, materials and wage bill in the census and that trade on at least one side, and I drop the exporters whose products are all Yarn-Forward. This leaves the trader panel of Table 16. Attrition is larger in 2018, and the wage-bill condition is responsible for most of it. The reason behind this is that before the 2021 subcontracting reform many manufacturers hired their workers through a separate staffing company, meaning that they reported no payroll of their own.

There is a problem with the CE 2024 extraction, where the imported-materials item is practically missing (2 importers, compared to 3,202 in 2019). This means that census materials for a 2024 importer already include what it imports. For the non-maquila customs importers of the 2023 panel I therefore compute domestic materials as census materials minus customs imports, with a floor at zero.

Figure 8: Sector cost shares γ_s across census years



Notes: γ_s by NAICS-3 over all CE manufacturing firms. Open circles correspond to CE 2019, filled circles to CE 2024 and diamonds to the pooled value used in both periods.

D Discrete-Choice Fit by Block

Table 17 reports the fit of sourcing and destination participation block by block, together with the share of firms for which the model reproduces the entire sourcing set. On the eight targeted margins the data and model shares coincide by construction of the inversion, which is why the informative columns are the "correct %" ones, given that these score the in/out status of each firm cell by cell. On the sourcing side the model gets between 78 and 93% of the firm-block cells right. To put this into perspective, a random assignment with the same shares would get 54 to 70% right in every block, and 76 to 77% in U.S. & CA, where participation is high, meaning that the model does better than this benchmark everywhere. On the destination side it gets 70 to 83% of the cells right, against 50 to 81% for the random benchmark.

E Estimated Firm Primitives: 2018

This appendix repeats for 2018 (5,726 traders) what Section 5.2 reports for 2023.

Table 17: Discrete-choice fit: sourcing and destination participation

	2018			2023		
Margin	Data %	Model %	Correct %	Data %	Model %	Correct %
Sourcing: share of firms sourcing from						
U.S. & CA	86.4	86.4	78.0	86.7	86.7	78.8
Asia excl. China	73.9	73.9	91.6	76.1	76.1	92.1
China	78.4	78.4	91.3	81.4	81.4	92.9
Europe	75.9	76.0	92.3	77.0	77.0	92.5
Rest of World	64.6	64.5	90.6	65.0	65.1	90.7
Destination: share of firms selling to						
U.S. & CA	78.0	78.0	76.5	79.8	79.8	79.2
Mexico	86.0	86.0	80.7	89.2	89.2	83.4
Rest of World	52.8	52.8	71.0	55.6	55.6	69.8
Exact sourcing set (% of firms)		61.8			61.8	

Notes: "Correct %" is the share of firms whose in/out status matches the data. Data and model shares coincide by construction of the inversion. MX-source participation is omitted, as it is 100% in the model by construction ($\kappa_s^{MX} \equiv 0$) against 71% (2018) and 67% (2023) of firms with positive domestic materials in the data.

E.1 Cross-Firm Distribution

Table 18 is the 2018 counterpart of Table 9. The sourcing efficiencies again have standard deviations of 2.0 to 2.1 log units, Mexico has the highest mean, the CH, EU and AX blocks are near a common lower mean and RoW is below them. Between the two periods the mean goes up for every sourcing-efficiency coordinate, by 0.03 to 0.18 log units, the most for China and the least for Mexico. On the demand side the mean imputed $\log b$ is 4.3 at U.S. & CA, 5.5 at Mexico and 3.3 at RoW, whereas among the firms observed in each market it is 5.5, 6.0 and 4.0.

Table 18: Estimated firm primitives: cross-firm distribution (2018)

Coordinate	mean	sd	observed
Sourcing efficiency $\log a_{f,s}$ *(U.S. & CA* $\equiv 0$*)*			
Mexico	−0.140	2.130	0.57
Asia excl. China	−1.267	2.039	0.70
China	−1.130	2.001	0.73
Europe	−1.050	2.018	0.71
Rest of World	−1.640	2.047	0.61
Market reach $\log b_{f,d}$			
U.S. & CA	5.259	1.810	0.78
Mexico	5.967	1.424	0.86
Rest of World	3.674	1.558	0.53

Notes: 5,726 traders. Construction as in Table 9.

E.2 Ownership Groups

Tables 19a and 19b are the 2018 counterparts of Tables 10a and 10b. The home-region pattern shows to be the same as in 2023. Mexican-owned firms have their highest sourcing efficiency in Mexico (+0.84), AX-owned firms in Asia excluding China (+1.68), Chinese-owned firms in China (+1.54) and EU-owned firms in Europe (+0.86), whereas U.S. & CA-owned firms are negative on every block other than their own, and once again the RoW-owned group does not peak at home. On the demand side, the group from each region has the highest mean market reach in that region's market, and in this period this also holds for the Rest of World, with Mexican-owned firms in Mexico (6.1), U.S. & CA-owned firms in the U.S. & CA market (6.4) and the remaining foreign owners in the Rest of World (4.0).

Table 19a: Estimated sourcing efficiency by ownership group (2018)

	Ownership group					
	Mexico	U.S. & CA	Asia excl. China	China	Europe	Rest of World
Firms	3,248	1,544	244	20	491	179
Mean sourcing efficiency $\log a_{f,s}$*, by source block (U.S. & CA* $\equiv 0$*)*						
Mexico	**0.84**	−2.20	0.42	−0.64	−0.34	−0.43
U.S. & CA	0.00	**0.00**	0.00	0.00	0.00	0.00
Asia excl. China	−1.80	−0.88	**1.68**	−0.46	−0.65	−0.71
China	−1.46	−0.85	−0.18	**1.54**	−0.49	−0.85
Europe	−1.18	−1.45	−1.13	−0.54	**0.86**	−0.48
Rest of World	−1.74	−1.86	−2.01	−1.43	−0.46	**−0.73**

Notes: Construction and groups as in Table 10a. Bold: each group's home block.

Table 19b: Estimated market reach by ownership group (2018)

	Ownership group		
	Mexico	U.S. & CA	Other foreign
Firms	3,248	1,544	934
Mean market reach $\log b_{f,d}$*, by destination*			
Mexico	**6.11**	5.82	5.72
U.S. & CA	4.72	**6.42**	5.21
Rest of World	3.43	3.97	**4.03**

Notes: Construction and groups as in Table 10b. Bold: each group's home market.

E.3 Variation Explained by Observables

Table 20 is the 2018 counterpart of Table 11, from the same regressions (27). The fully interacted observables explain at most 46% of the cross-firm variation in any coordinate (Mexican sourcing efficiency, 0.461). Ownership is again the stronger observable for the sourcing-efficiency coordinates other than China, and sector is the stronger one for the three demand coordinates.

Table 20: Share of the cross-firm variation in $\log a_{f,s}$ and $\log b_{f,d}$ explained by ownership and sector (2018)

		R^2 from fixed effects for				Adding firm size $\log L_f$	
Latent	Coordinate	Own	Sector	Own + Sector	Own × Sector	Alone	+ Own × Sector
$\log a$	MX	0.377	0.192	0.444	0.461	0.019	0.462
	AX	0.148	0.145	0.233	0.248	0.037	0.265
	CH	0.046	0.136	0.157	0.172	0.022	0.187
	EU	0.093	0.057	0.137	0.155	0.045	0.195
	RoW	0.041	0.034	0.075	0.094	0.048	0.135
$\log b$	U.S. & CA	0.161	0.370	0.446	0.454	0.071	0.497
	MX	0.013	0.225	0.240	0.248	0.031	0.292
	RoW	0.032	0.265	0.287	0.296	0.045	0.332

Notes: 5,726 traders. Construction as in Table 11.

E.4 Correlation Structure

Figure 9 is the 2018 counterpart of Tables 12a and 12b in the main text. The correlation structure is stable across periods. For instance, the AX–CH sourcing-efficiency correlation, which I think of as an East-Asian bloc effect, is 0.53 in 2018 and 0.51 in 2023, and the destination correlation between U.S. & CA and RoW is 0.50 and 0.52. I take this as evidence that these correlations are a structural feature of the trader panel and not something specific to one period. Part of the stability is built in, though, since both periods come from the same pooled imputation surface.

F Shapley Decomposition

This appendix completes the description of the decomposition of Section 6.1. The ten factors are listed in Table 21.

Weights. Since $|\mathcal{J}|!\,(m-|\mathcal{J}|-1)!/m! = 1/\left[m\binom{m-1}{|\mathcal{J}|}\right]$, the weight in (28) is the probability of ending up with one particular sub-model with $|\mathcal{J}|$ switched inputs when the size of the

Figure 9: Correlation structure of estimated firm primitives (2018)

	MX	AX	CH	EU	RoW
MX	1.00				
AX	−.10	1.00			
CH	−.07	.53	1.00		
EU	.19	.34	.30	1.00	
RoW	.09	.28	.29	.47	1.00

(a) Sourcing efficiency $a_{f,s}$

	U.S. & CA	MX	RoW
U.S. & CA	1.00		
MX	.27	1.00	
RoW	.50	.40	1.00

(b) Market reach $b_{f,d}$

Notes: 5,726 traders, 2018. Panels show the lower triangle of the correlation matrix (upper triangle omitted by symmetry). Panel (a) omits U.S. & CA ($\log a_{f,\text{U.S. \& CA}} \equiv 0$).

Table 21: Factors of the Shapley decomposition

Group	Factor	2018 value → 2023 value
Policy	R	Rules of origin (μ_p^{CTC}, μ_p^{RVC}): NAFTA to USMCA rule file
Firm	$\hat{a}, \hat{b}$	Re-estimated primitives of the continuing firms (4,185 establishments present in both panels): their 2023 sourcing efficiencies a_f, market reach b_f, cost shares γ_s, product sets and imputed cells, inverted conditional on the 2023 aggregates
	E	Entrants: the 963 firms present only in the 2023 panel are added
	X	Exiters: the 1,541 firms present only in the 2018 panel are removed
Aggregates	ψ	Country sourcing shifters ψ_s
	κ	Locations of the fixed-cost distributions (μ_s, μ_d)
	D	Foreign demand (E_d, P_d), $d \in \{\text{U.S. \& CA}, \text{RoW}\}$
	A_o	Pinned wage, 165.4 to 168.7 thousand 2023 pesos per paid worker
	$\bar{L}$	Labor force, 55.6 to 60.6 million
	Θ	Domestic demand

Notes: 5,726 firms in the 2018 panel, 5,148 in the 2023 panel. When the three firm factors are at their 2023 values the firm set is the 2023 panel. Every firm keeps its own fixed-cost draw in all solves.

sub-model is drawn first, all sizes equally likely, and then one sub-model of that size is drawn at random (Audoly et al., 2025). Another way to see C_k is as the average, over the $m!$ orders in which the factors could be switched, of the change in X at the step in which k is switched. This is why no order has to be chosen. Having to choose one is the path-dependence problem of sequential decompositions (Fortin et al., 2011). The one-at-a-time channels of a sequential decomposition, where each factor is switched alone from E_A, are the $\mathcal{J} = \emptyset$ terms of (28), each with weight $1/m$.

Pairwise form. To evaluate (28) one needs $X(\mathcal{J})$ for all $2^m = 1{,}024$ subsets. What I use instead is

$$C_k \;=\; \tfrac{1}{2}\big[X(\{k\}) - X(\emptyset)\big] \;+\; \tfrac{1}{2}\big[X(\mathcal{K}) - X(\mathcal{K} \setminus \{k\})\big], \tag{36}$$

which takes $2m + 2 = 22$ solves, $X(\emptyset)$, $X(\mathcal{K})$, each factor switched on alone and each factor switched off alone. In order to see when (36) and (28) coincide, one can write $X(\mathcal{J})$ as the sum of the own effect of each factor plus the interactions among the factors in $\mathcal{J}$. The first bracket in (36) then corresponds to the own effect of k, whereas the second one is its own effect plus every interaction that k takes part in. Their average gives k its own effect plus half of every interaction it is in. If all interactions are between two factors, this is exactly (28), as each pairwise interaction gets split evenly between its two members. Interactions among three or more factors break this, because the halves over-count them, and the gap $X_B - X_A - \sum_k C_k$ is the measure of that part of the change. In Table 13 the gap is never above 0.3 of a point.

Solves. All 22 cells solve the same combined panel of the two periods. What changes from cell to cell is who is in it and with which data. A continuing firm uses its 2018 row, product set and fixed-cost draw or its 2023 ones, depending on the state of the $(\hat{a}, \hat{b})$ factor. Whether entrants and exiters are in the panel depends on the state of E and X. Each firm keeps the same draw in all cells, so κ only shifts where the distributions are located. With every factor at its 2018 value the cell reproduces E_A, and with every factor at its 2023 value it reproduces E_B.